

Image Restoration

Modelling the image degradations, inverting the model

Acquired image is a degraded version of the original scene

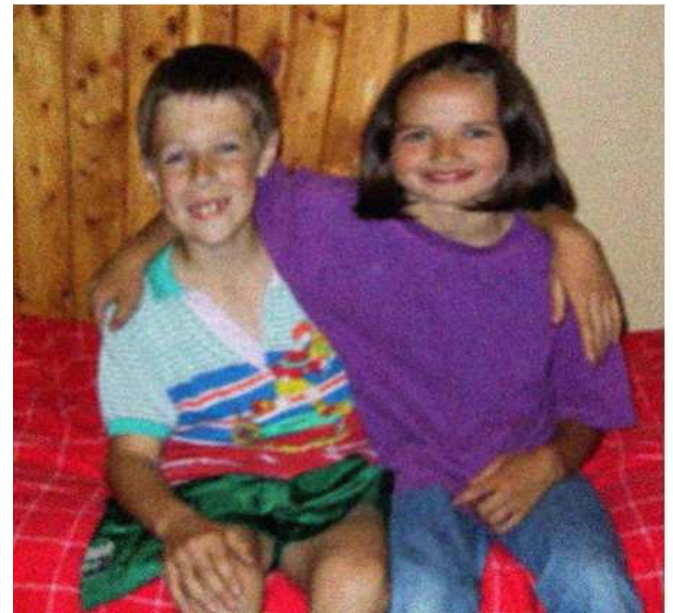


Image degradation model

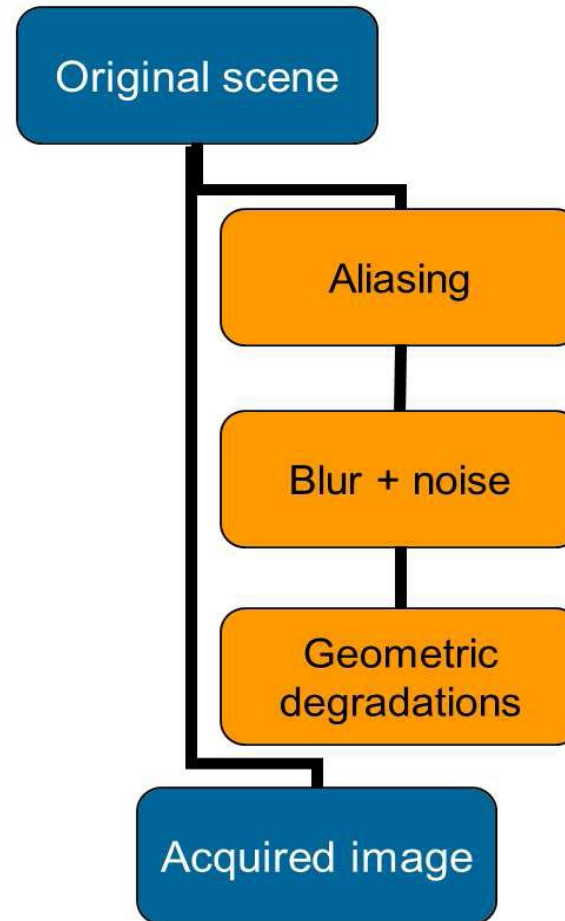
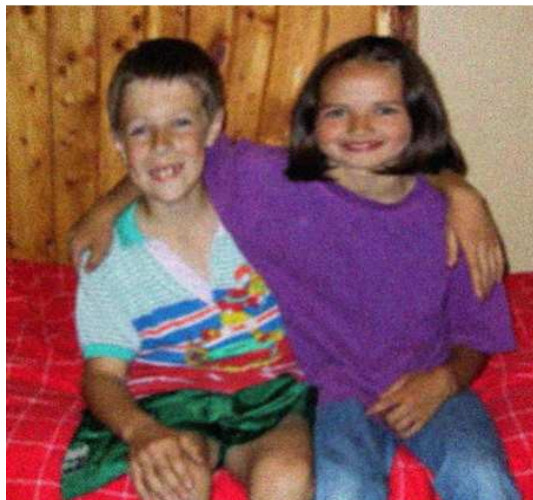
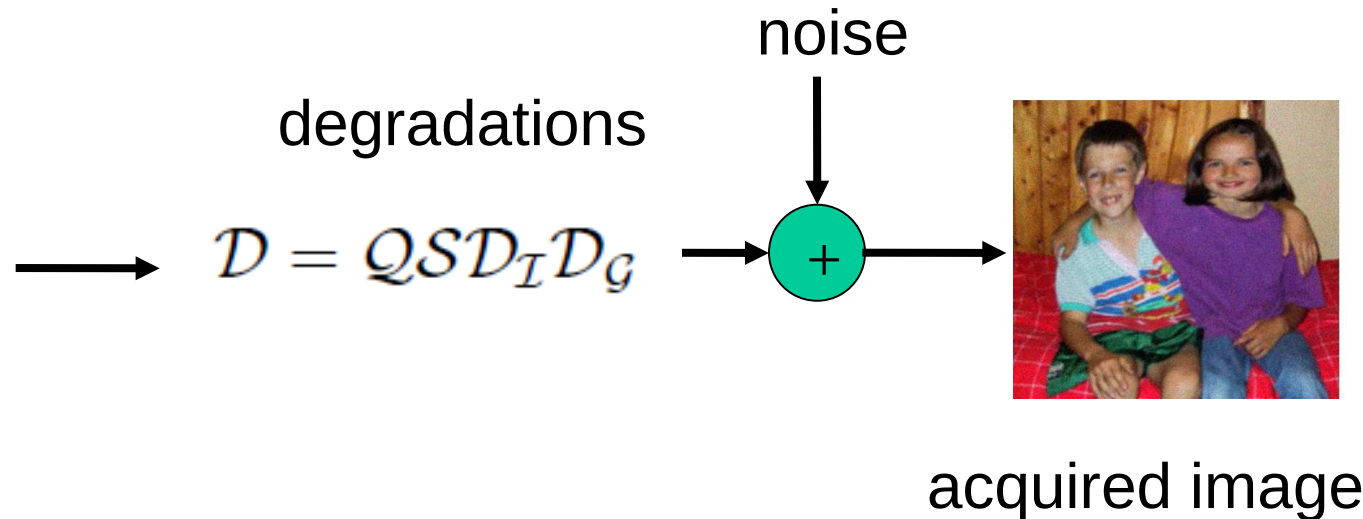


Image acquisition model



original scene



acquired image

$$\mathcal{D}_G(f)(x, y) = f(\tau(x, y))$$

$$\mathcal{D}_I(f)(x, y) = \int_{\Lambda} \int_T \int \int h(x, y, a, b, \lambda, t) f(a, b) da db d\lambda dt$$

Sources, models and appearance of individual degradations

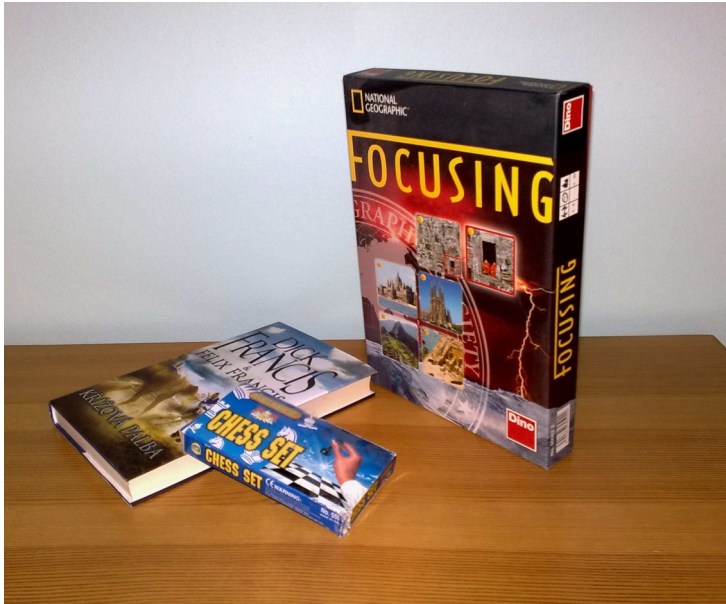
Geometric deformation

$$\mathcal{D}_G(f)(x, y) = f(\tau(x, y))$$



Image spatial transformation/warping
Invertible in most cases

Non-ideal view, planar objects $\rightarrow$ projective geometry



$$u = \frac{a_0 + a_1x + a_2y}{1 + c_1x + c_2y}$$

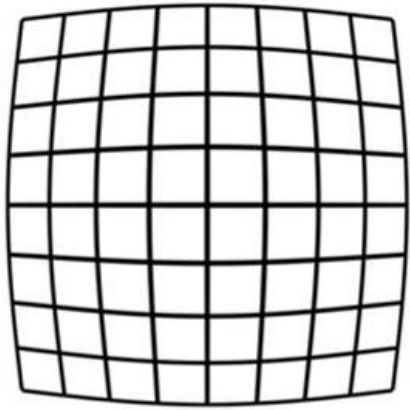
$$v = \frac{b_0 + b_1x + b_2y}{1 + c_1x + c_2y}$$

2D projection of curved surface $\rightarrow$ nonlinear models

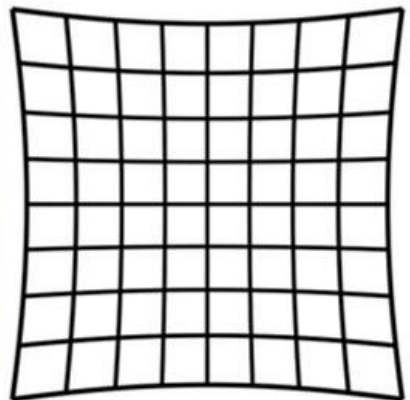


$$\mathcal{D}_{\mathcal{G}}(f)(x, y) = f(\tau(x, y))$$

Lens distortion



Barrel



Pincushion



Non-standard lenses/cameras (omnivision, fish-eye)



Dealing with geometry

- Modelling and inverting D



- Comparing two images of the same scene regardless of D

Image Registration

Overlaying two or more images of the same scene

Image registration

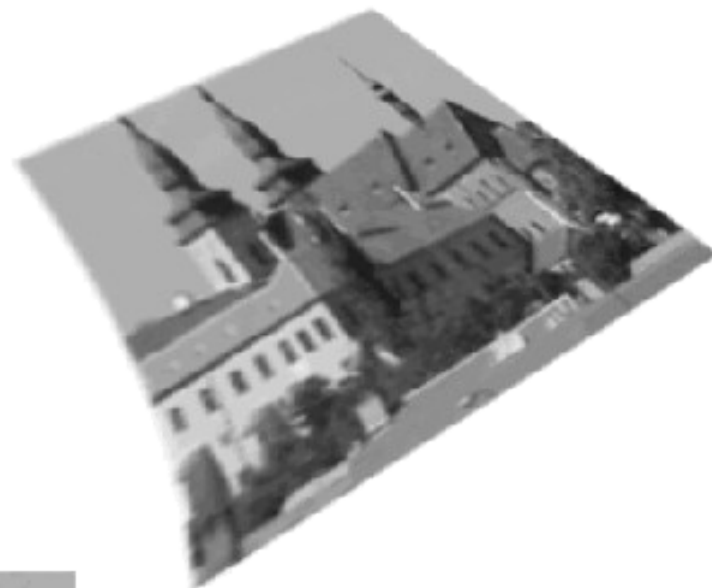


IMAGE REGISTRATION CATEGORIES



- 1. Different viewpoints - multiview**
- 2. Different times - multitemporal**
- 3. Different modalities - multimodal**
- 4. Scene to model registration**

IMAGE REGISTRATION METHODOLOGY

Four basic steps of image registration

1. Control point selection



1. Control point selection

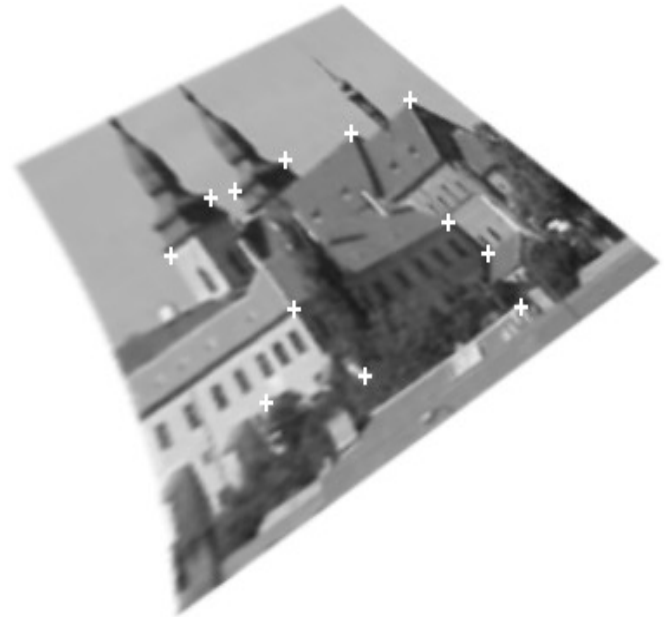


IMAGE REGISTRATION METHODOLOGY

Four basic steps of image registration

1. Control point selection

2. Control point matching



2. Control point matching

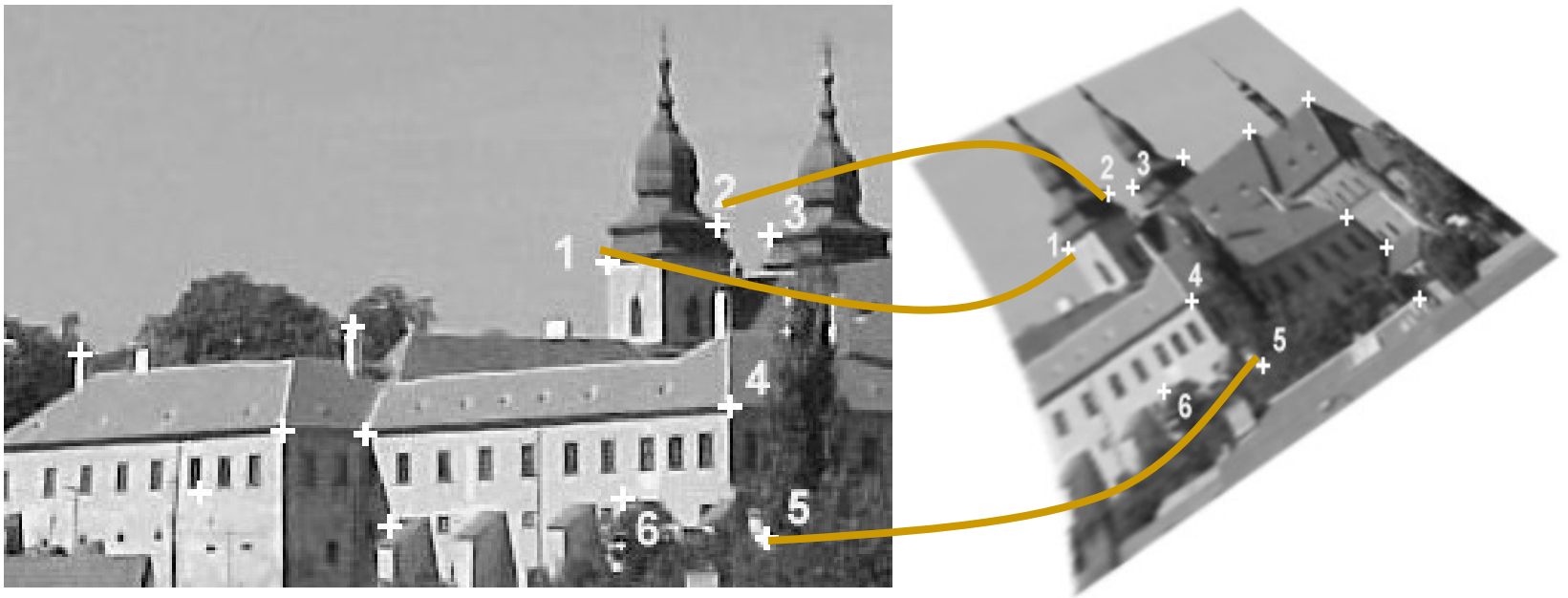


IMAGE REGISTRATION METHODOLOGY

Four basic steps of image registration

1. Control point selection
2. Control point matching
3. Transform model estimation



3. Mapping function design

$$u = f(x,y)$$

$$v = g(x,y)$$

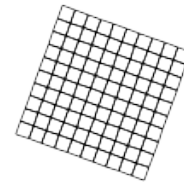
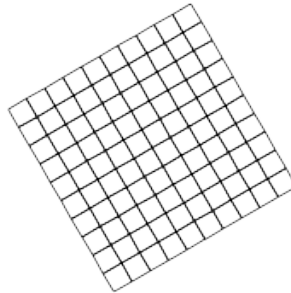
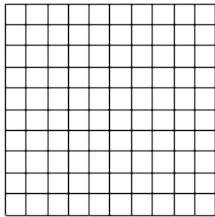
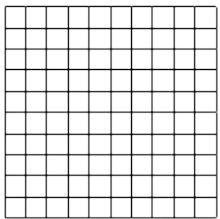


IMAGE REGISTRATION METHODOLOGY

Four basic steps of image registration

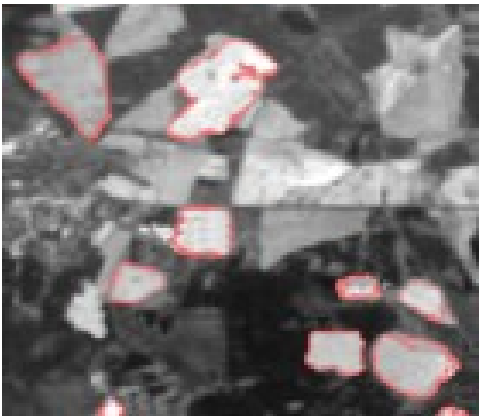
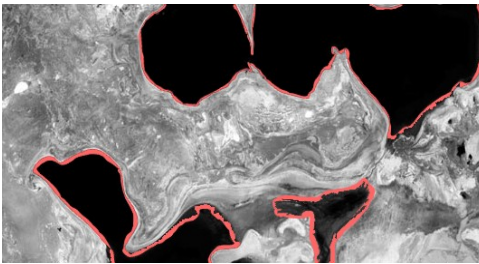
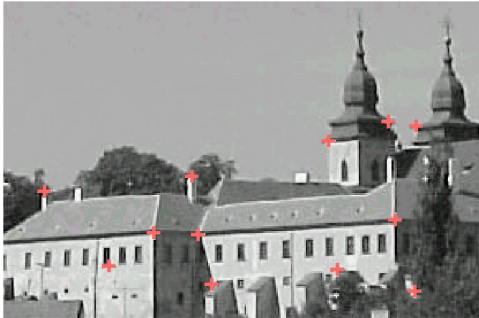
1. Control point selection
2. Control point matching
3. Transform model estimation
4. Image resampling and transformation



4. Image resampling and transformation

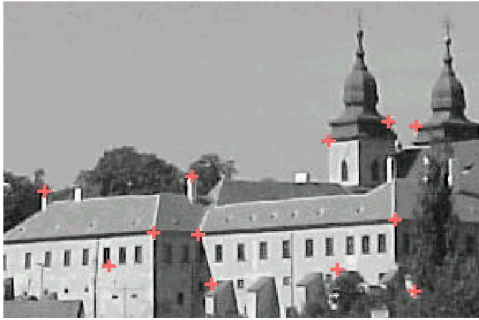


CONTROL POINT SELECTION



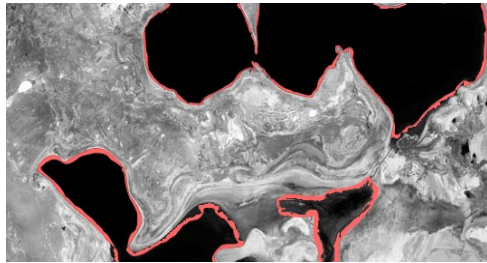
- distinctive points
- corners
- lines
- closed-boundary regions
- virtual invariant regions
- window centers

DESIRABLE PROPERTIES OF CONTROL POINTS

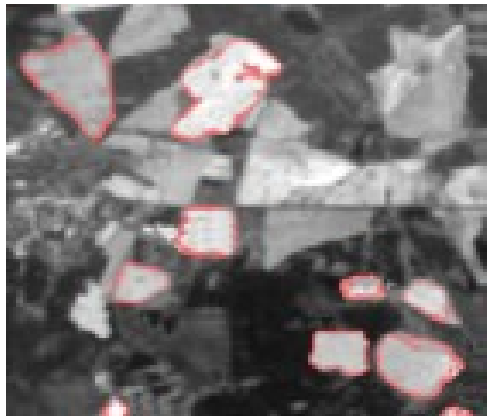


Distinctive and detectable objects

Physical interpretability



Frequently spread over the image



Enough common elements in all images

Robust to degradations

Control point matching

Signal-based methods

Similarity measures calculated directly from the image graylevels

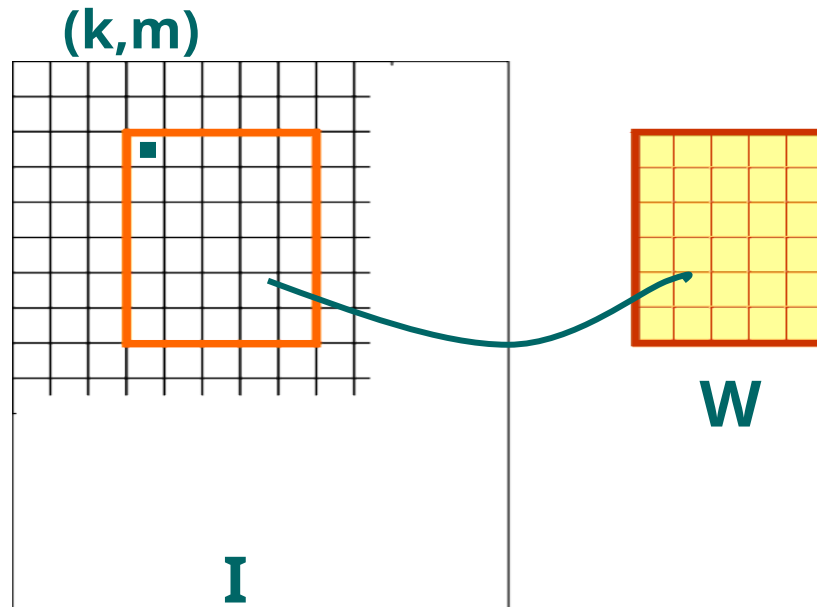
Examples - Image correlation, image differences, phase correlation, mutual information, ...

Feature-based methods

Symbolic description of the features

Matching in the feature space
(classification)

Image correlation



$$C(k,m) = \frac{\sum (I_{k,m} - \text{mean}(I_{k,m})) \cdot (W - \text{mean}(W))}{\sqrt{\sum (I_{k,m} - \text{mean}(I_{k,m}))^2} \cdot \sqrt{\sum (W - \text{mean}(W))^2}}$$

Image correlation

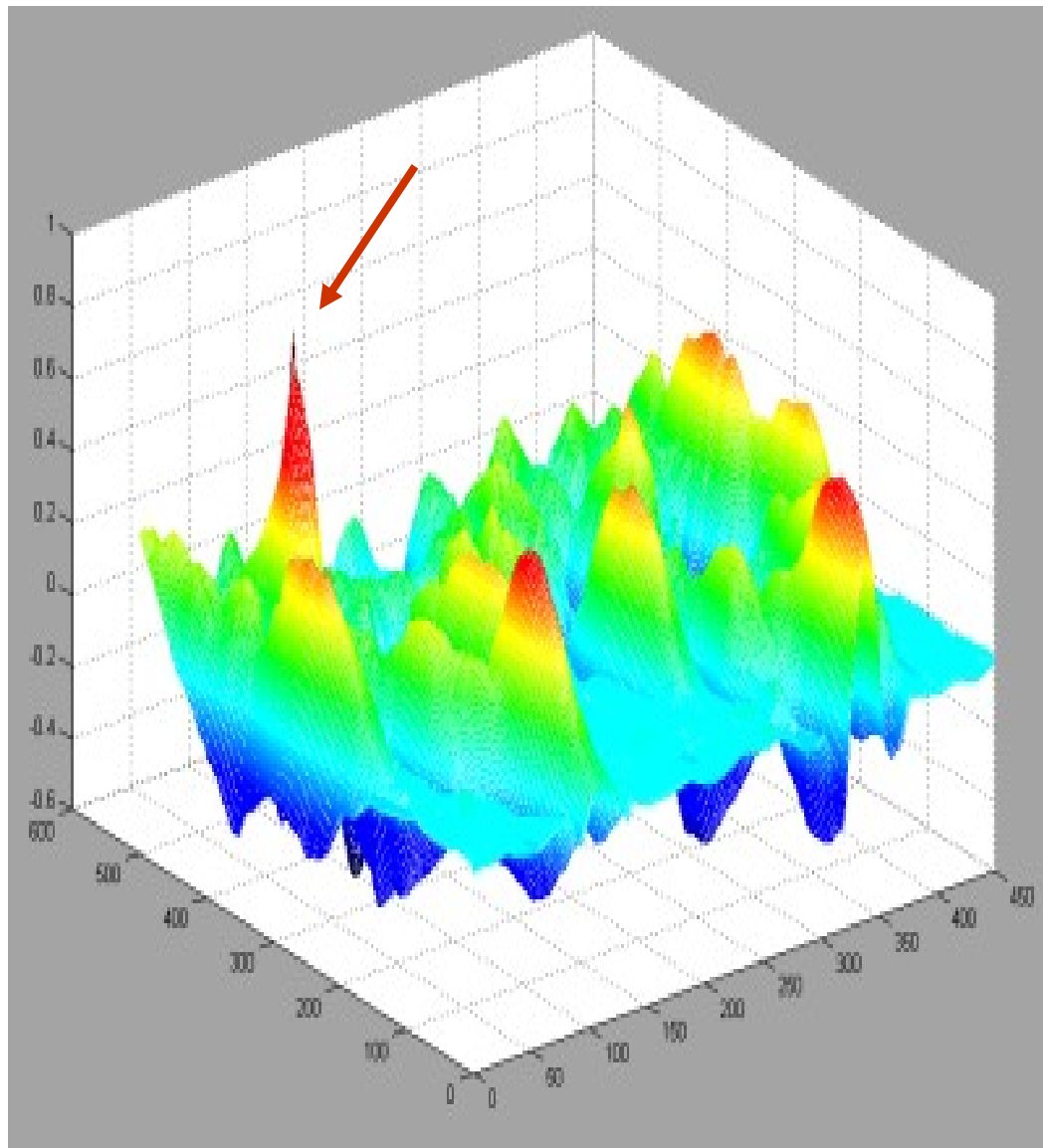


Image correlation

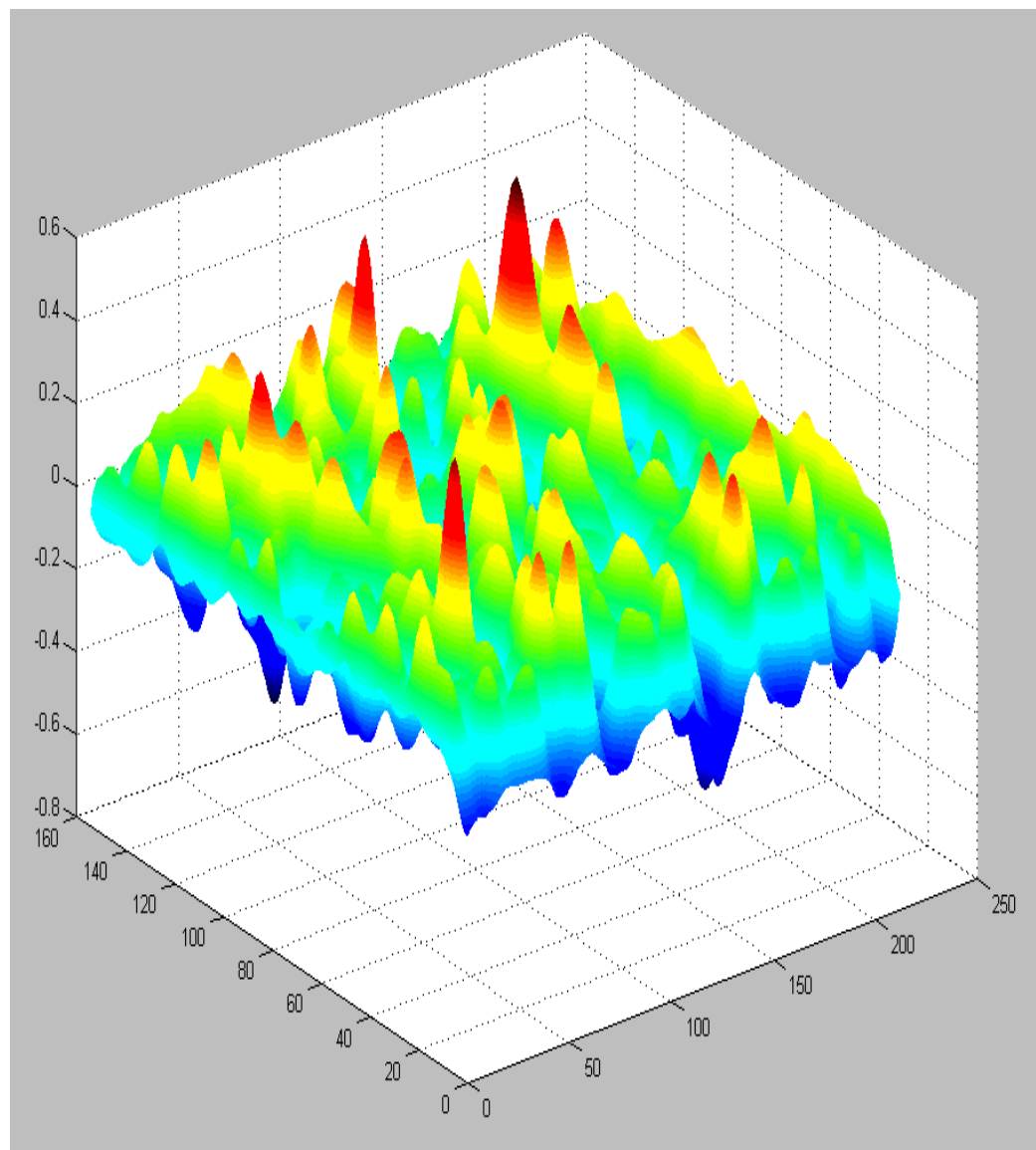
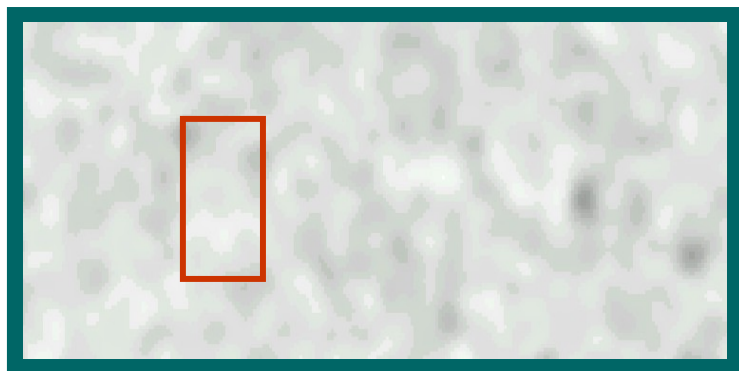
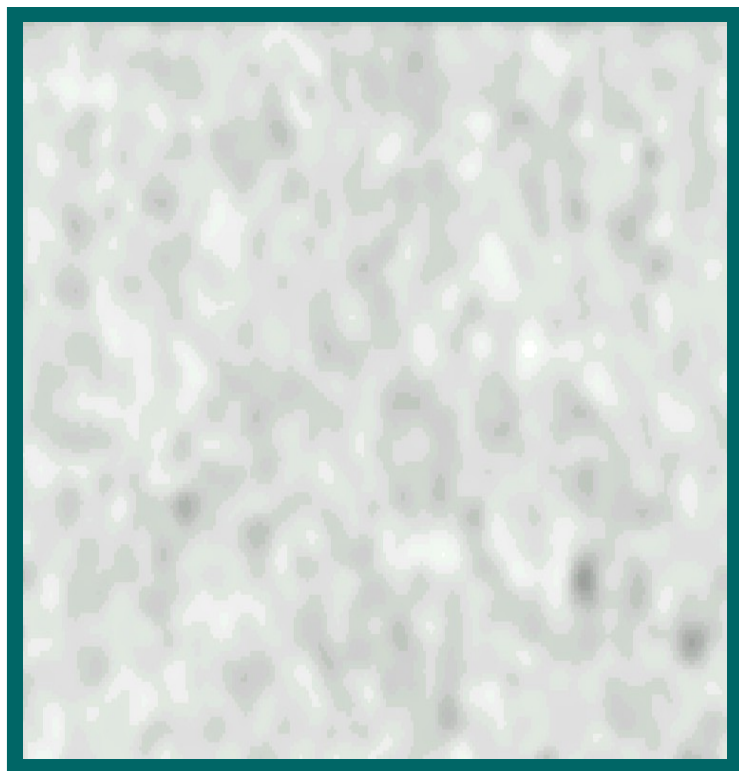
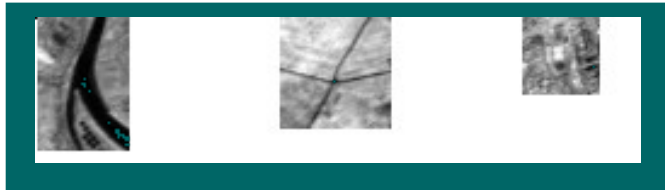


Image correlation – template matching



Band-to-band registration



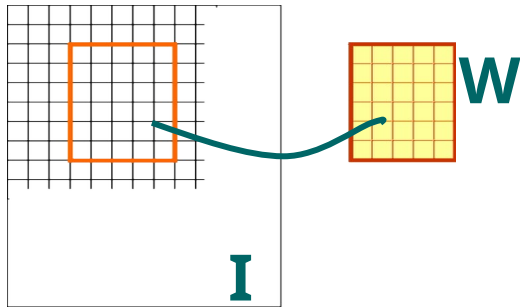
5



7



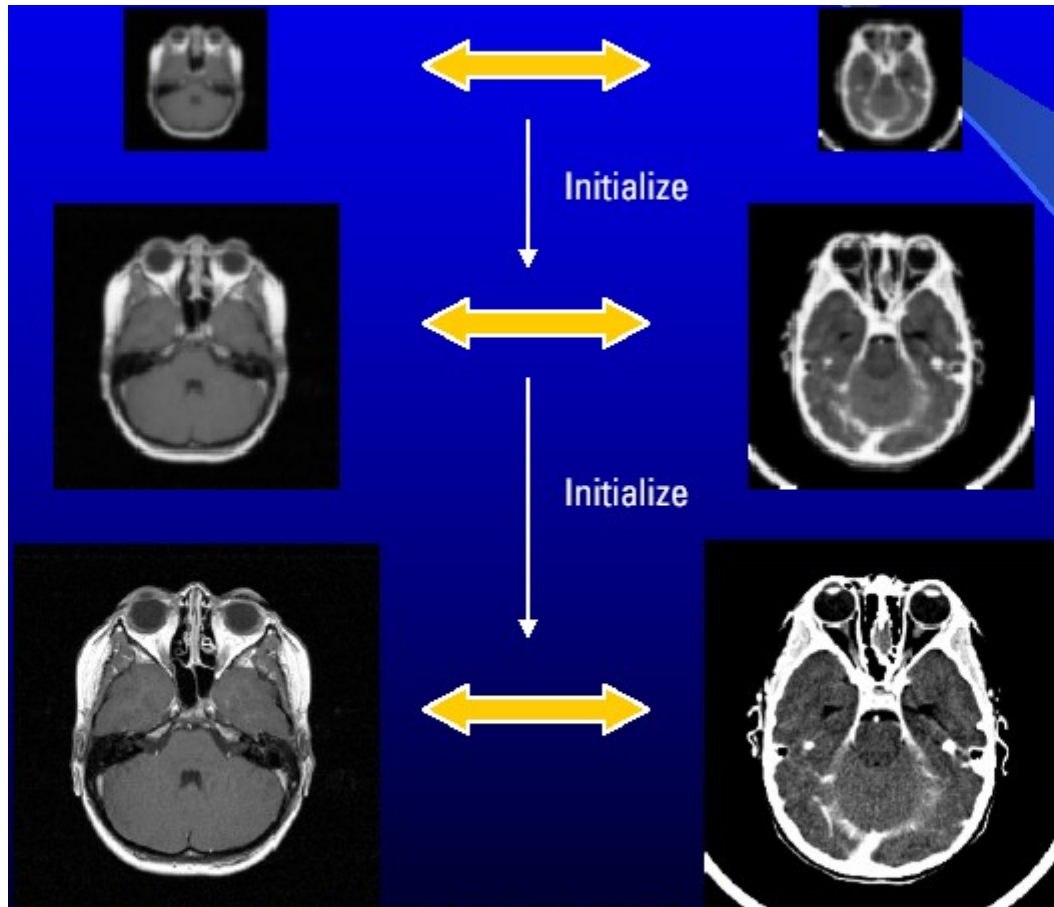
Correlation-like methods



- edge correlation, vector correlation
 - correlation in F domain (phase corr)
 - other similarity measures (L1 norm, Mutual information)
 - extension to more complex transformations (rotation, scaling)
 - extension to 3-D
 - subpixel accuracy
 - speed-up techniques (optimization algorithms, pyramidal representation, real-time HW implementation)
-

Pyramidal representation

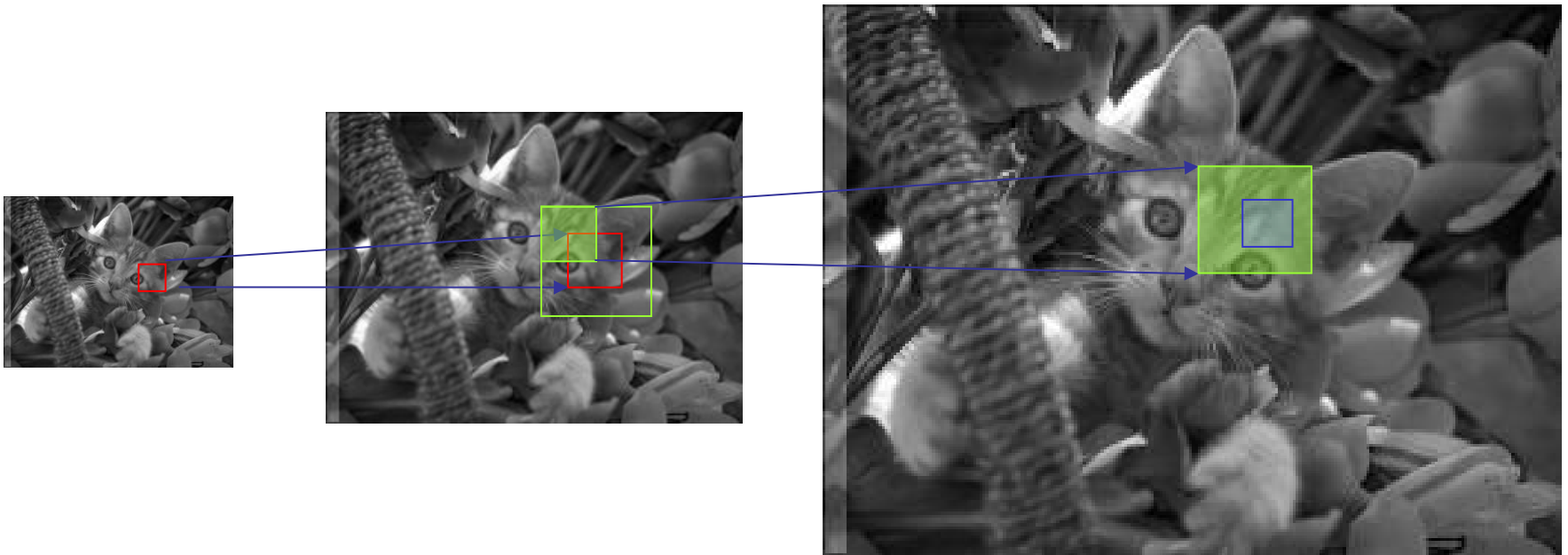
Processing from coarse to fine level



FEATURE MATCHING

PYRAMIDAL REPRESENTATION

processing from coarse to fine level



wavelet transform

Phase correlation

Equivalent to standard correlation of “whitened” images (similar to correlation of edges).

Does not depend on actual image colors.



Phase correlation

Fourier Shift Theorem:

If $f(x)$ is shifted by a to $f(x-a)$ then the FT magnitude stays constant while the phase is shifted by $-2\pi au$.

Shift parameter can be detected by a comparison of both spectra and by Inverse Fourier Transform.

Phase correlation

Cross-power spectrum

$$\frac{W \cdot F^*}{|W \cdot F|} = e^{-2\pi i (ua + vb)}$$

F - Fourier transform of f

W - Fourier transform of w

a, b - unknown shift parameters

$*$ - complex conjugate

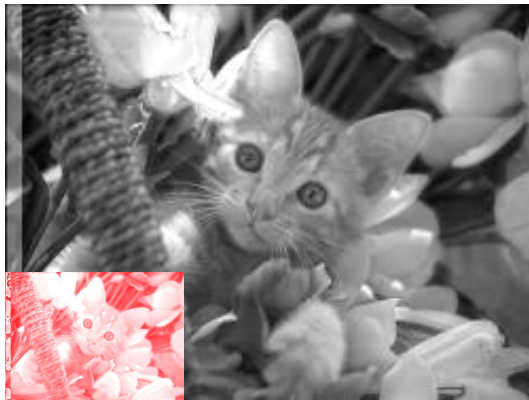
$$\text{IFT} (e^{-2\pi i (ua + vb)}) = \delta(x-a, y-b)$$

Image f



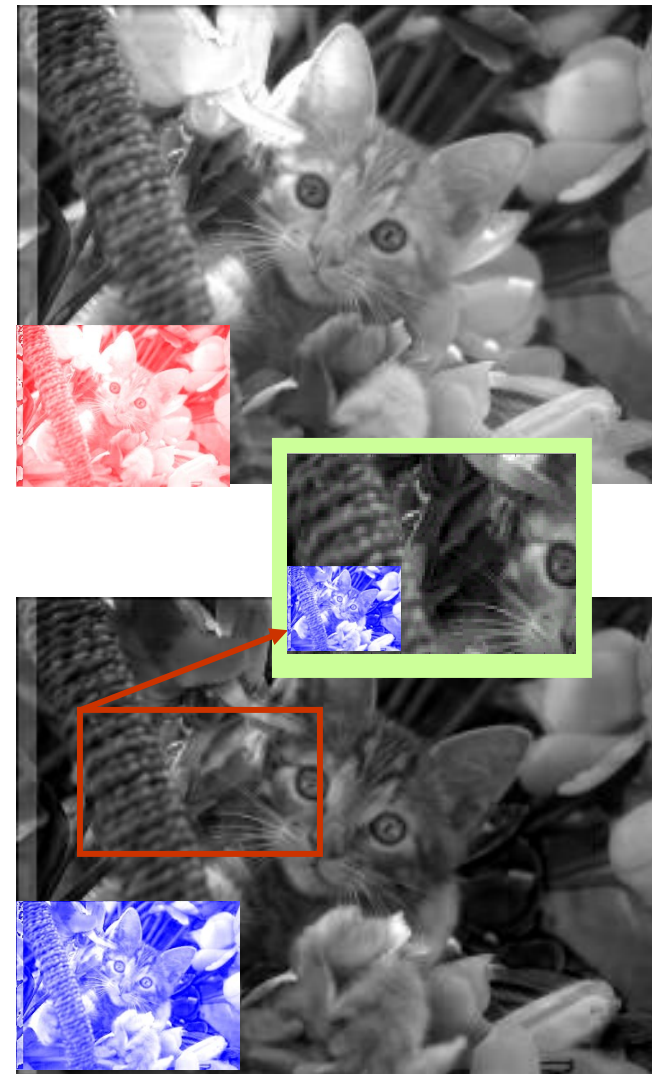
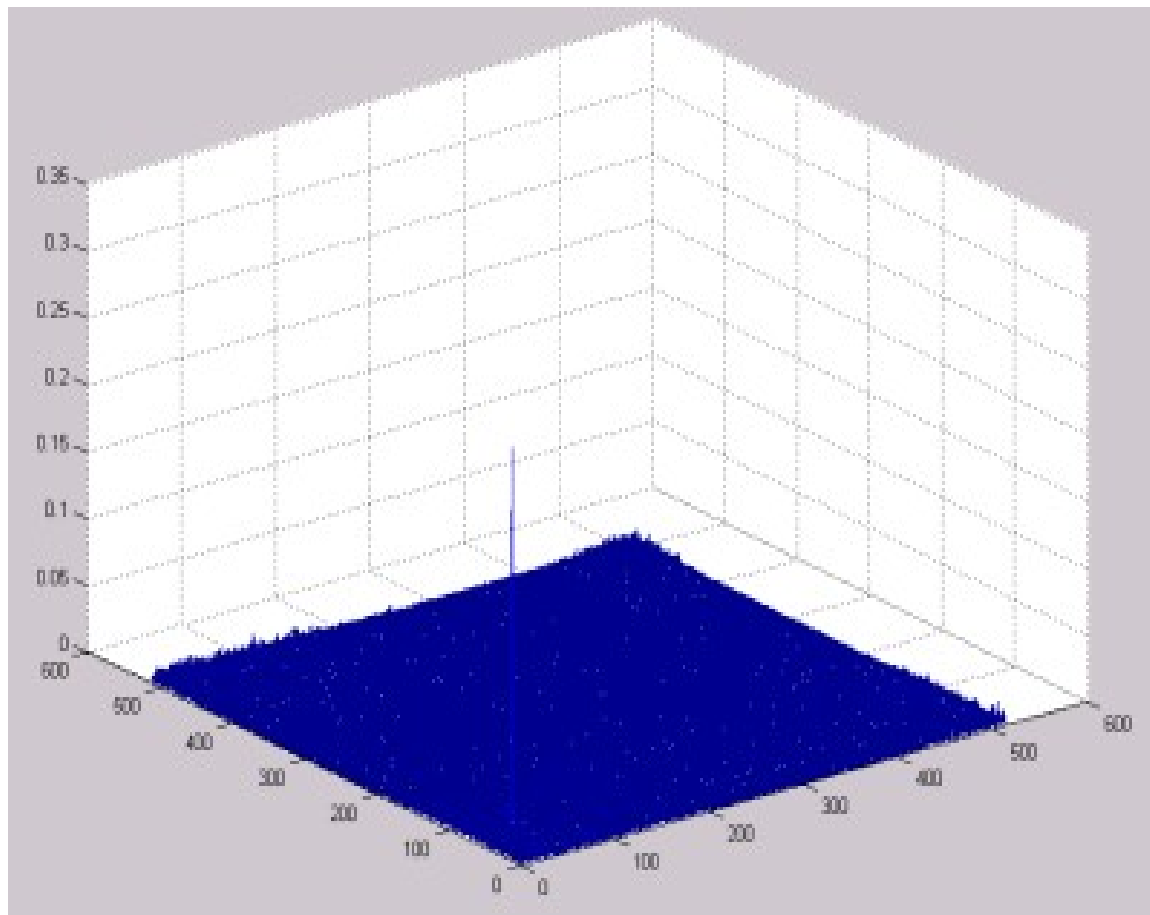
Window w

- **Multimodal registration - experiment**
- **Comparison of image correlation and phase correlation**



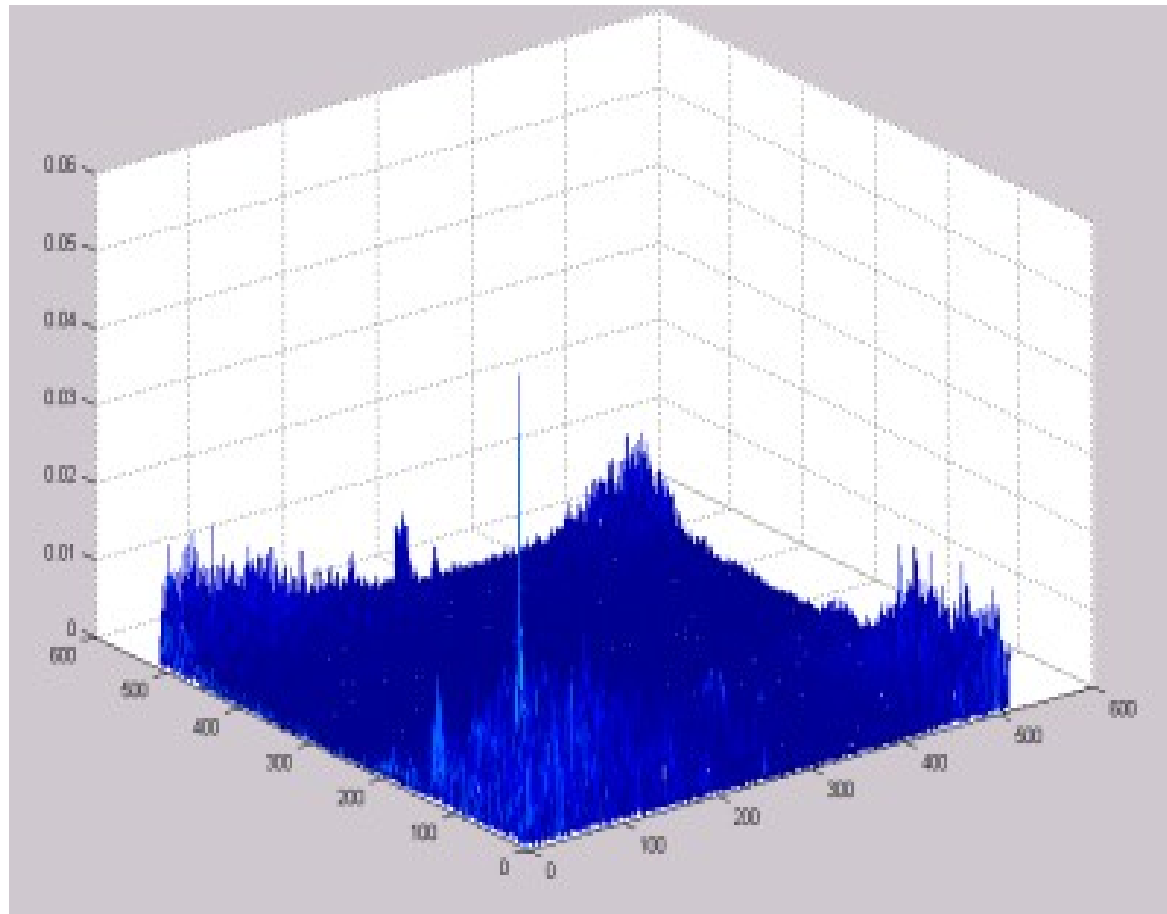
Channel to channel registration

phase-correlation $R-B$



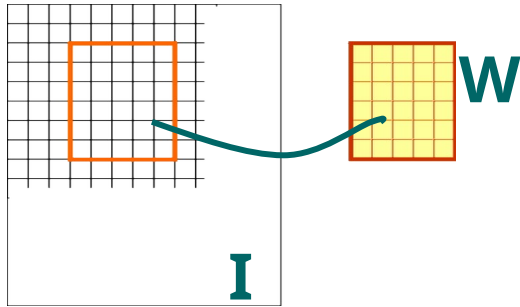
Multimodal registration

phase-correlation IR-IR

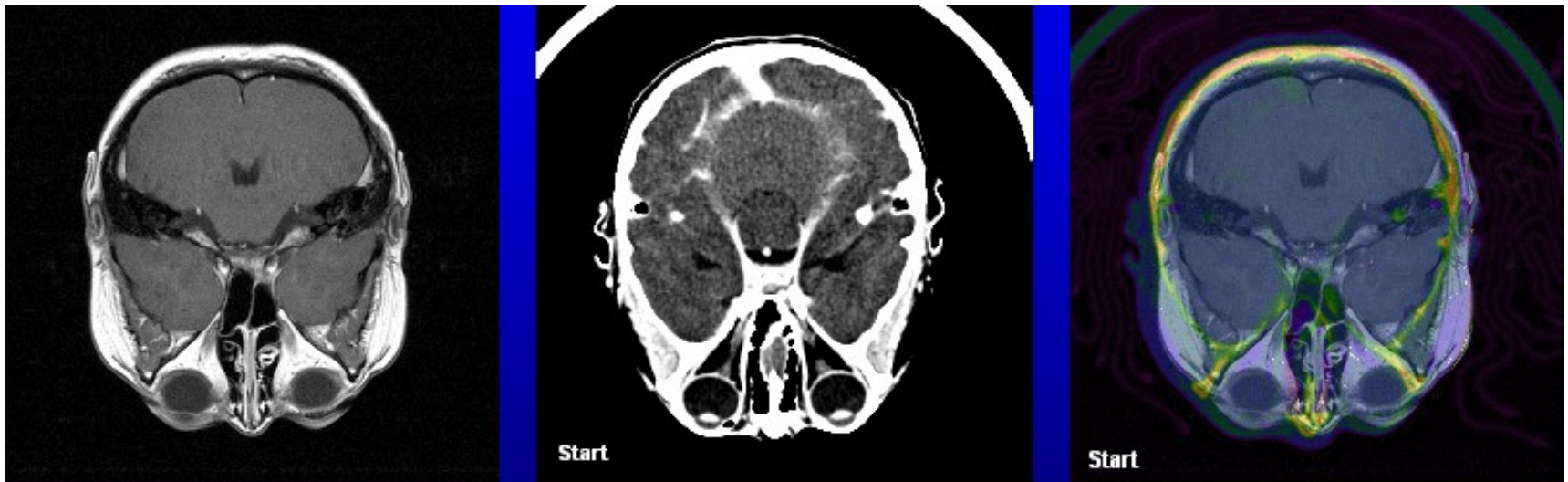


Mutual information method

Statistical measure of the dependence between two images



$$MI(f,g) = H(f) + H(g) - H(f,g)$$



MUTUAL INFORMATION

Entropy

$$H(X) = - \sum_x p(x) \log p(x)$$

Joint entropy

$$H(X, Y) = - \sum_x \sum_y p(x, y) \log p(x, y)$$

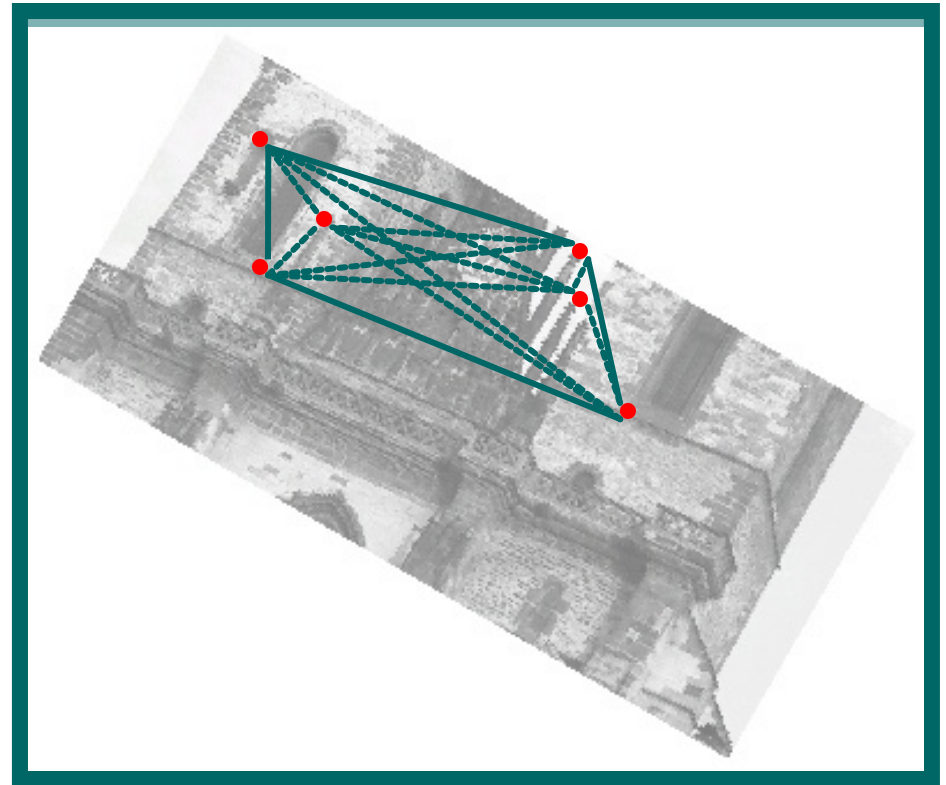
Mutual information

$$I(X; Y) = H(X) + H(Y) - H(X, Y)$$

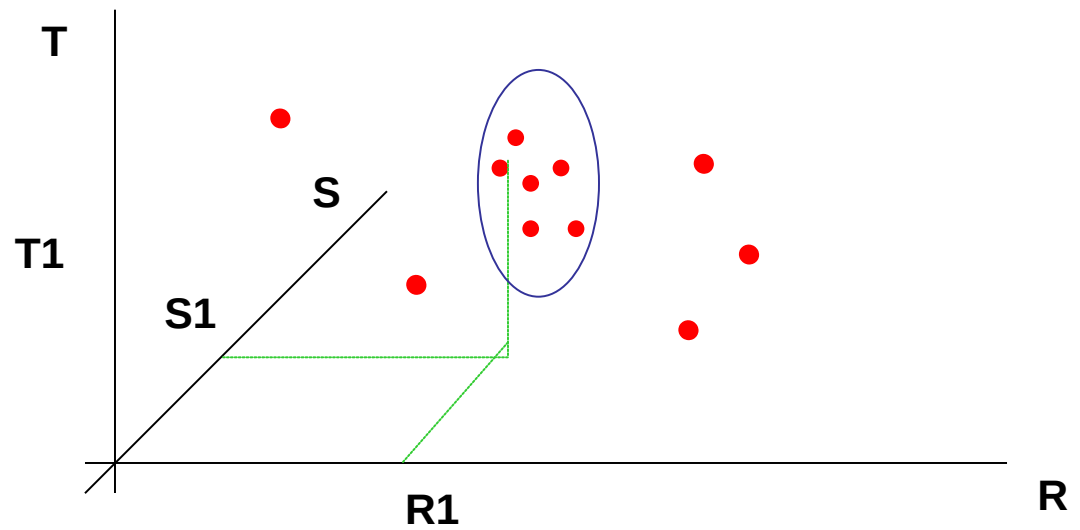
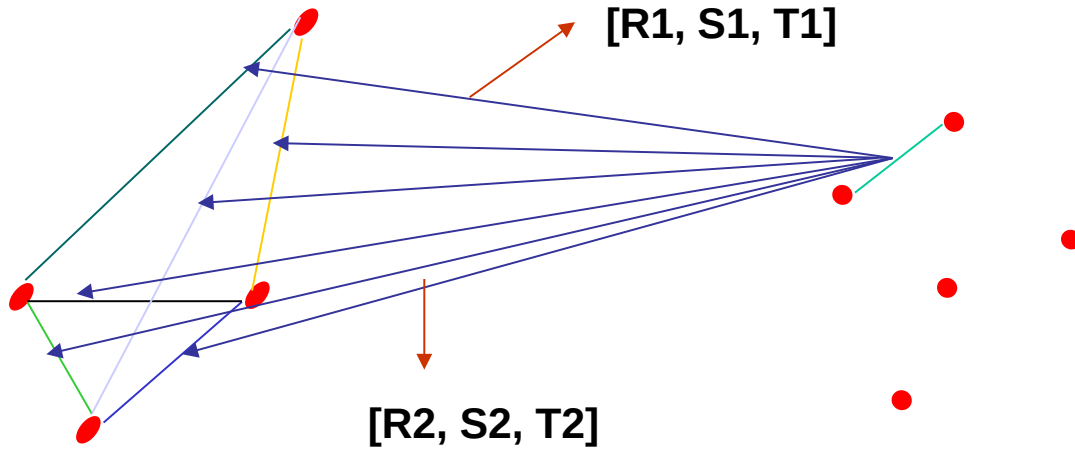
Feature-based methods

- **Combinatorial matching (no feature description). Graph matching, parameter clustering. Global information only is used.**
- **Matching in the feature space (pattern classification). Local information only is used.**
- **Hybrid matching (combination of both to get higher robustness)**

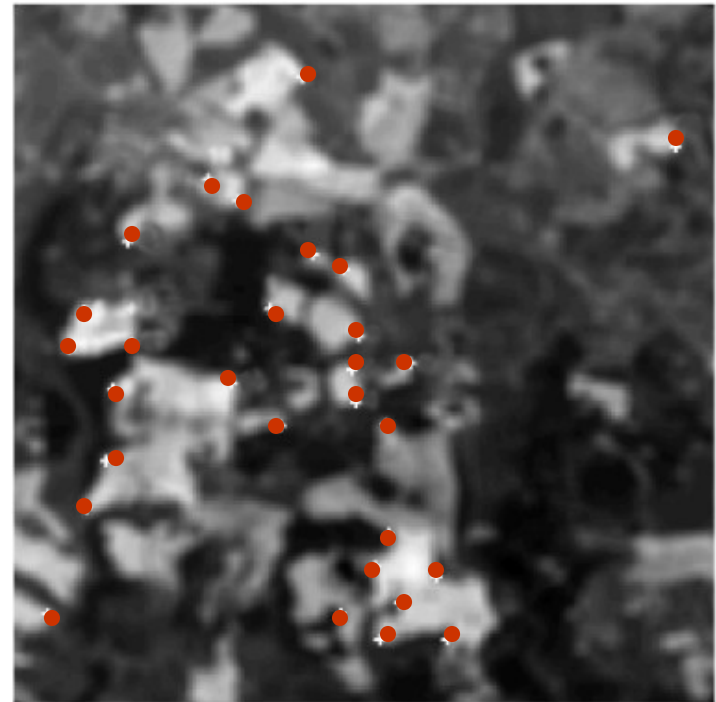
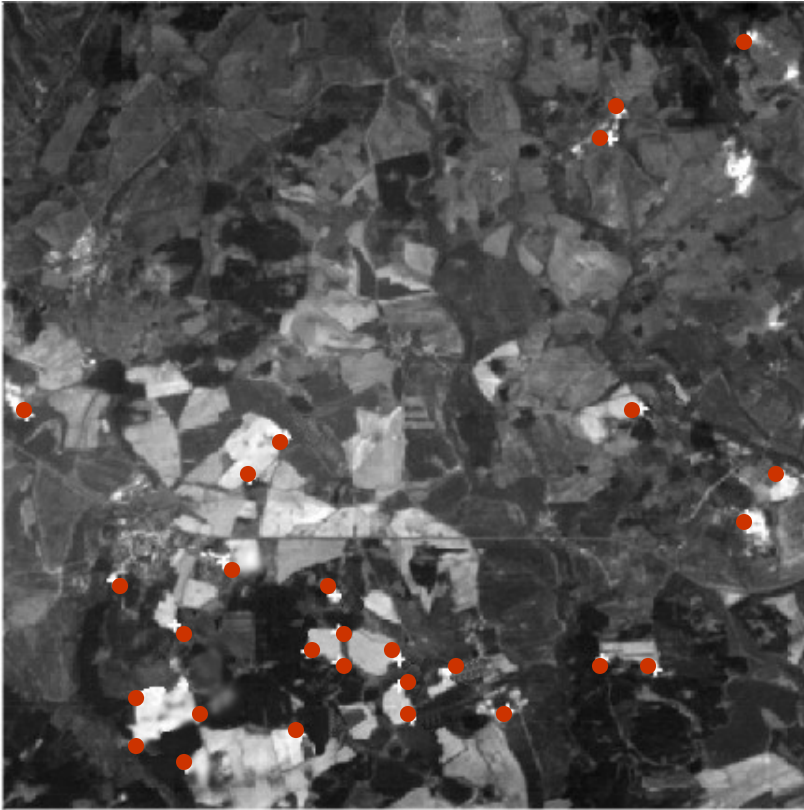
Combinatorial matching



PARAMETER CLUSTERING



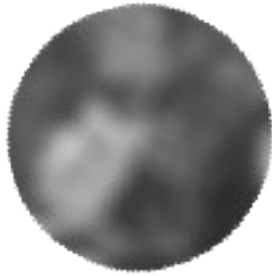
Matching in the feature space



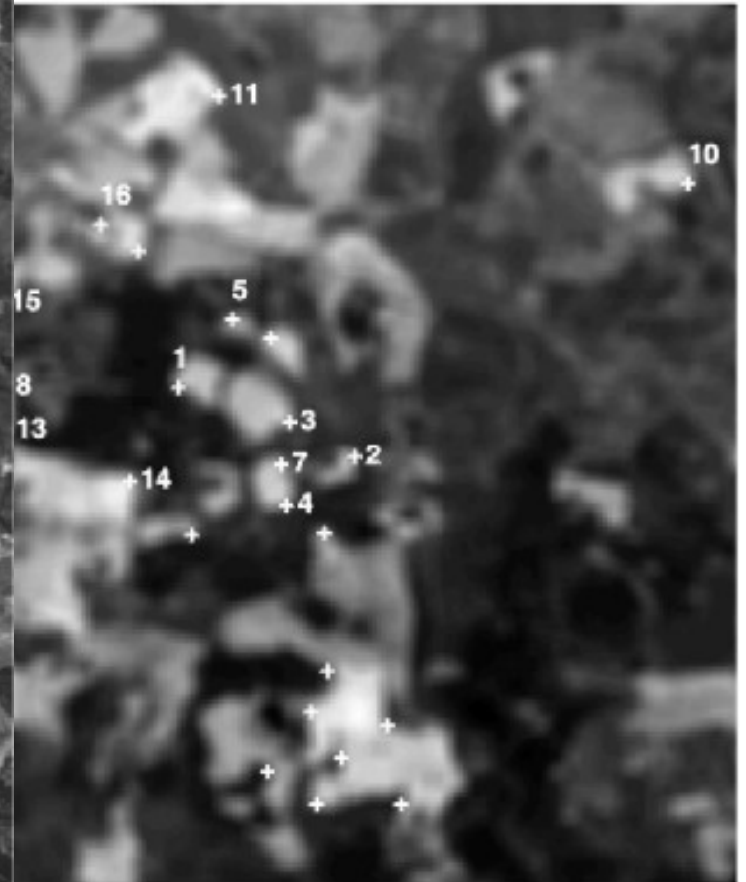
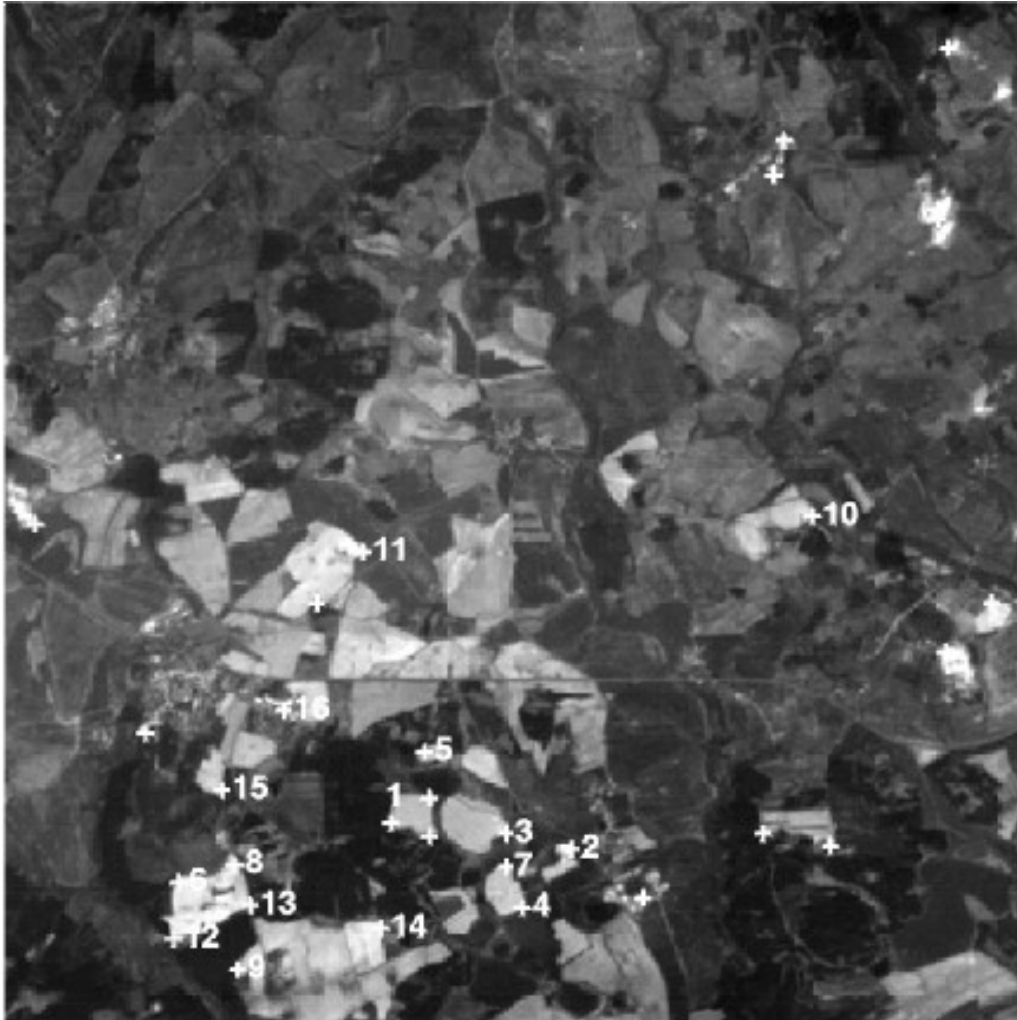
$$\min_{k,m} \text{distance}((v1_k, v2_k, v3_k, \dots), (v1_m, v2_m, v3_m, \dots))$$

Features for CP description

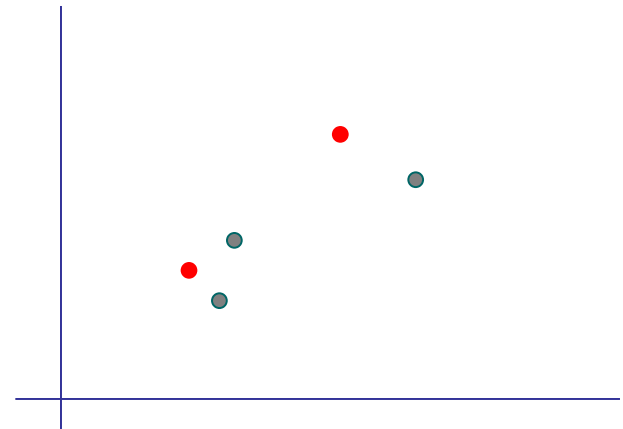
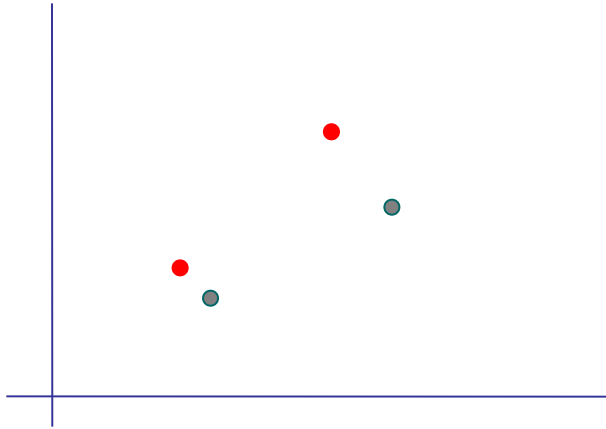


- **calculated over a circular neighborhood of each CPC**
 - **invariant to all assumed image degradations**
- 
- **discriminative**
 - **robust to noise**

Matching result



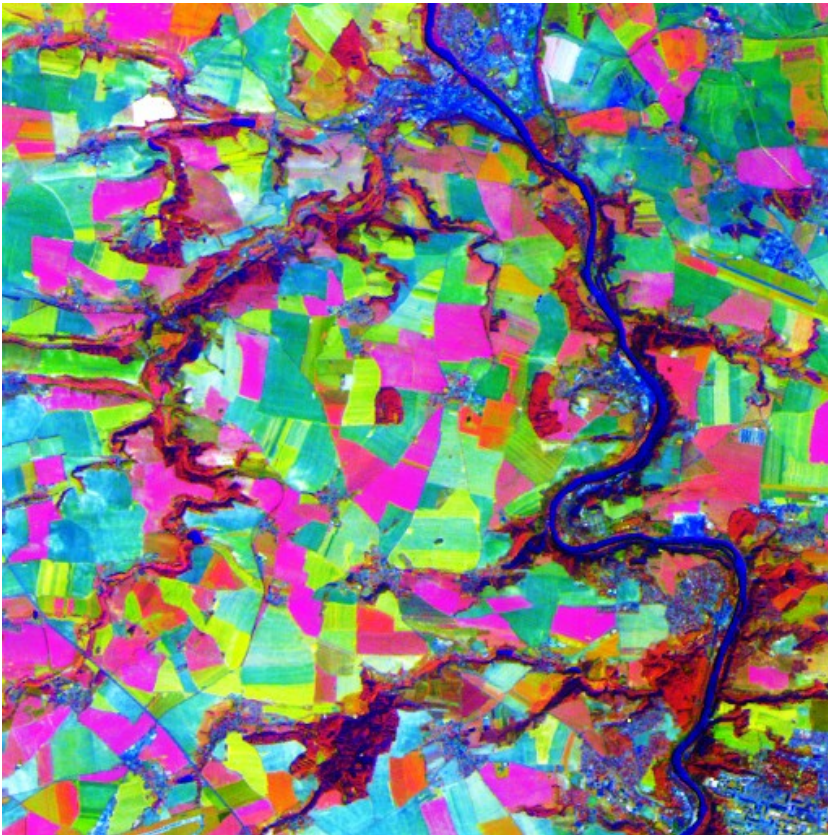
ROBUST MATCHING IN THE FEATURE SPACE



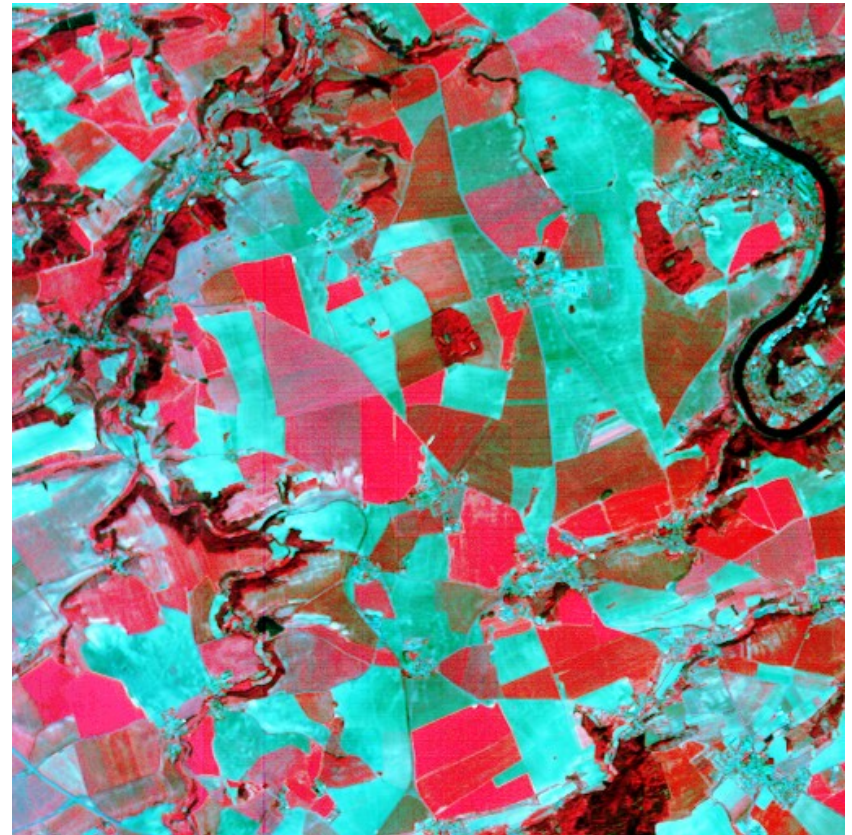
Hybrid matching

- **Step 1: Matching few CP's by invariant features**
- **Step 2: Matching the rest of the CP's in the image space**

Hybrid matching

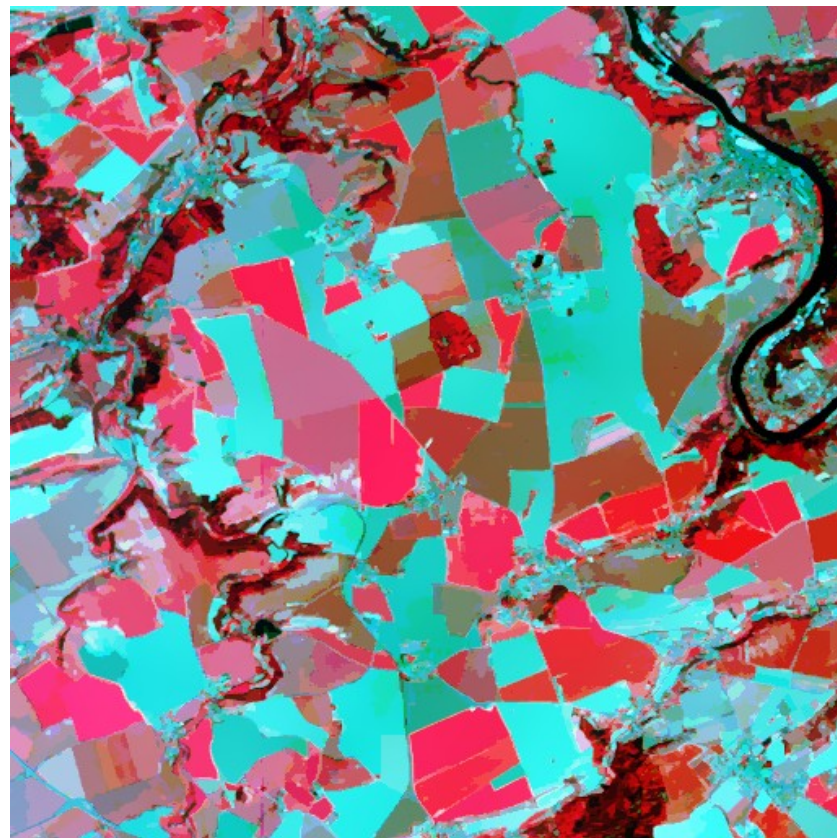
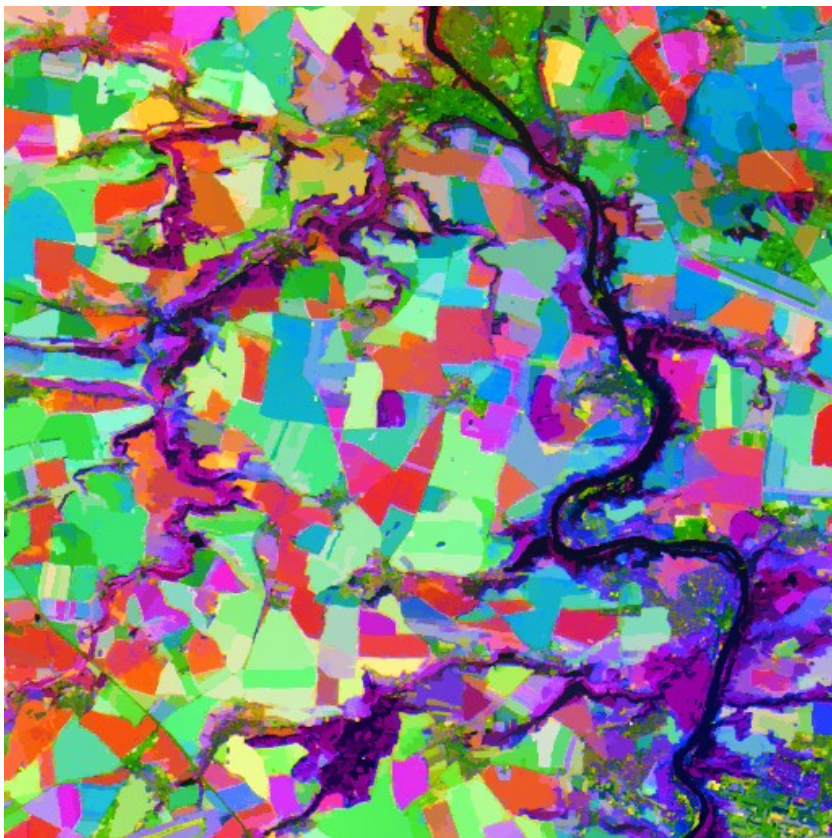


Landsat

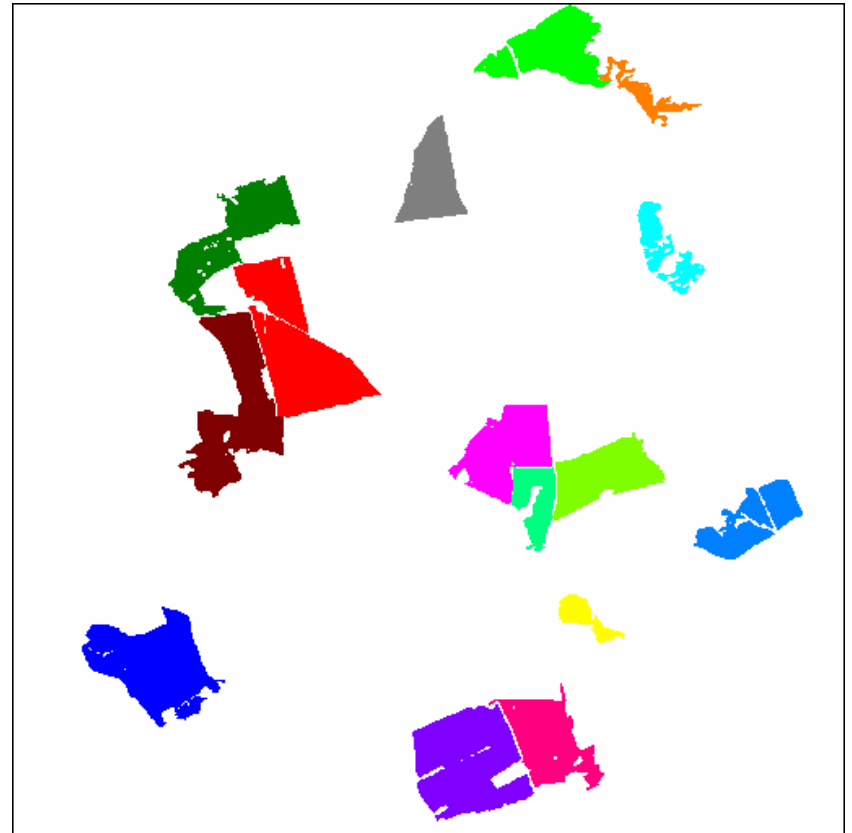
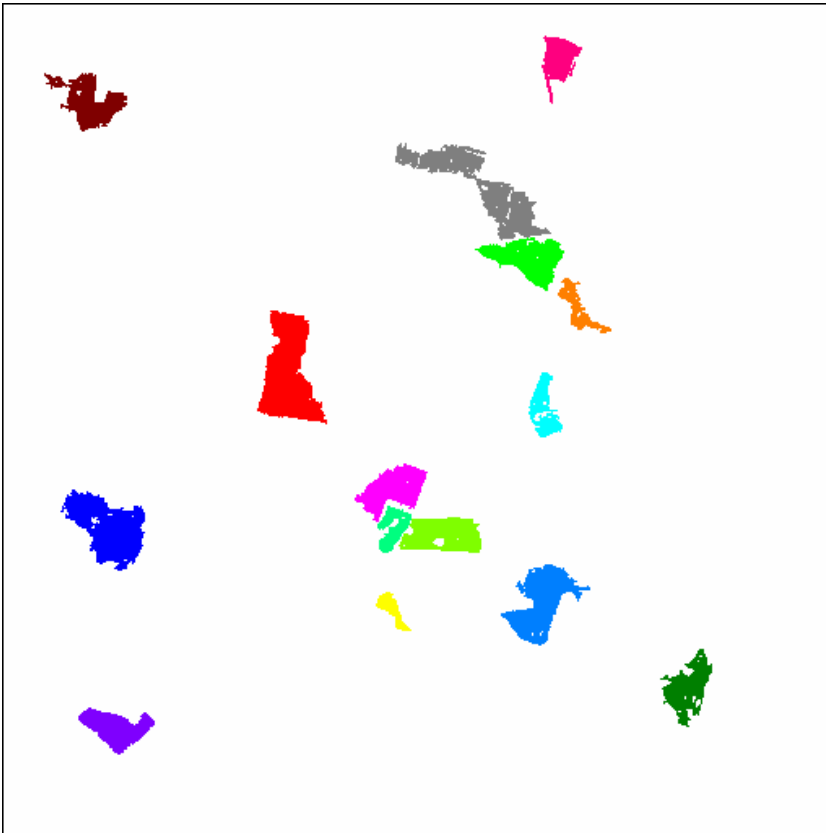


Spot

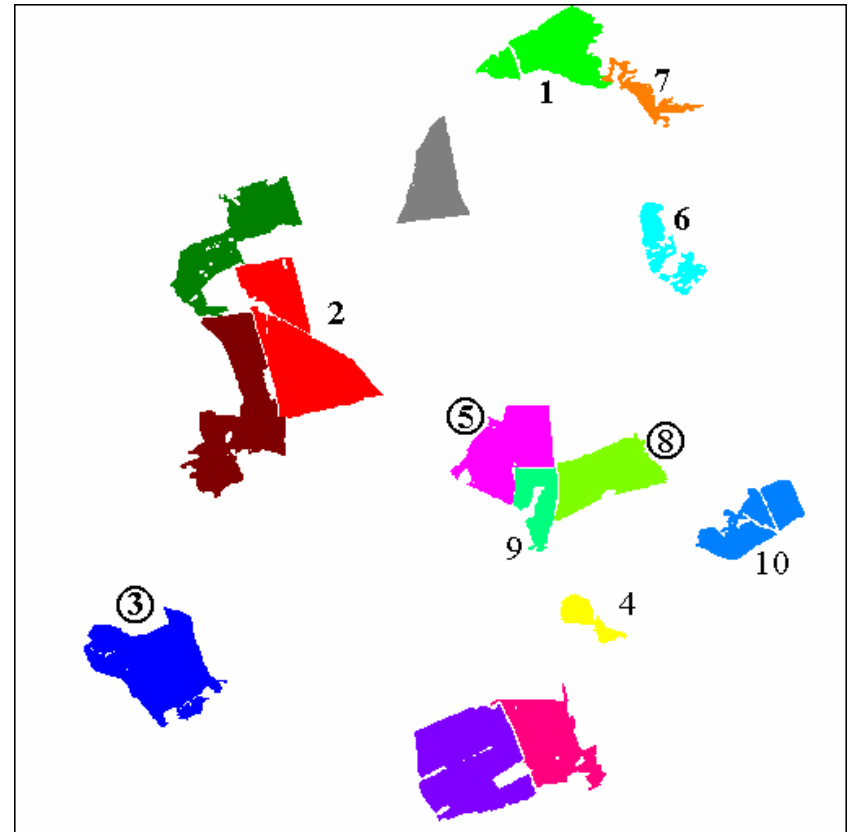
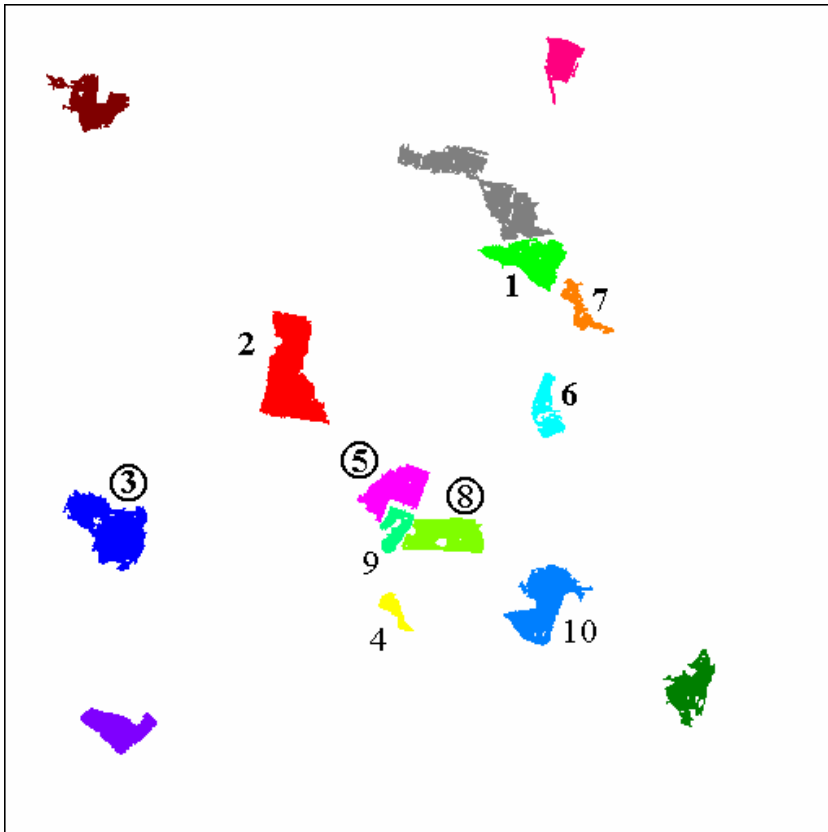
Segmentation by Mumford - Shah method



Selected regions



Matching pairs



Mapping function design

- **Global functions**

Similarity, affine, projective transform

Low-order polynomials

Mapping function design

- **Global functions**

Similarity, affine, projective transform

Low-order polynomials

- **Local functions**

Piecewise affine, piecewise cubic

Thin-plate splines

Radial basis functions

Global mapping functions

- Affine transform

$$x' = a_0 + a_1x + a_2y$$

$$y' = b_0 + b_1x + b_2y$$



Similarity transform – Least square fit

translation $[\Delta x, \Delta y]$, rotation φ , uniform scaling s

$$\begin{aligned} x' &= s (x \star \cos \varphi - y \star \sin \varphi) + \Delta x \\ y' &= s (x \star \sin \varphi + y \star \cos \varphi) + \Delta y \end{aligned}$$

$$s \cos \varphi = a, \quad s \sin \varphi = b$$

$$\min (\sum_{i=1} \{ [x'_i - (ax_i - by_i) - \Delta x]^2 + [y'_i - (bx_i + ay_i) - \Delta y]^2 \})$$

$$\begin{vmatrix} \Sigma(x_i^2 + y_i^2) & 0 \\ 0 & \Sigma(x_i^2 + y_i^2) \\ \Sigma x_i & -\Sigma y_i \\ \Sigma y_i & \Sigma x_i \end{vmatrix} \begin{vmatrix} \Sigma x_i & \Sigma y_i \\ -\Sigma y_i & \Sigma x_i \end{vmatrix} \begin{vmatrix} a \\ b \end{vmatrix} \begin{vmatrix} \Delta x \\ \Delta y \end{vmatrix} = \begin{vmatrix} \Sigma(x'_i x_i - y'_i y_i) & \Sigma(y'_i x_i - x'_i y_i) \\ \Sigma x'_i & \Sigma y'_i \end{vmatrix}$$

Global mapping functions

- Affine transform

$$x' = a_0 + a_1x + a_2y$$

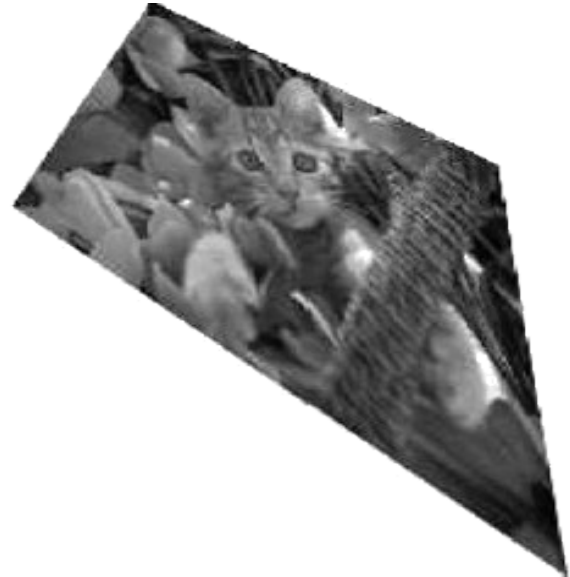
$$y' = b_0 + b_1x + b_2y$$



- Projective transform

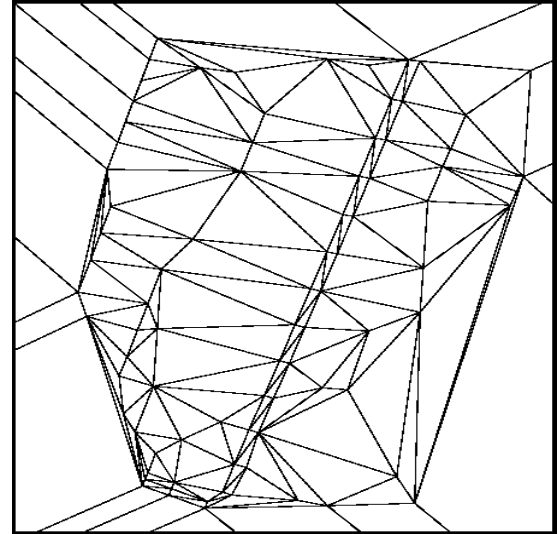
$$x' = (a_0 + a_1x + a_2y) / (1 + c_1x + c_2y)$$

$$y' = (b_0 + b_1x + b_2y) / (1 + c_1x + c_2y)$$



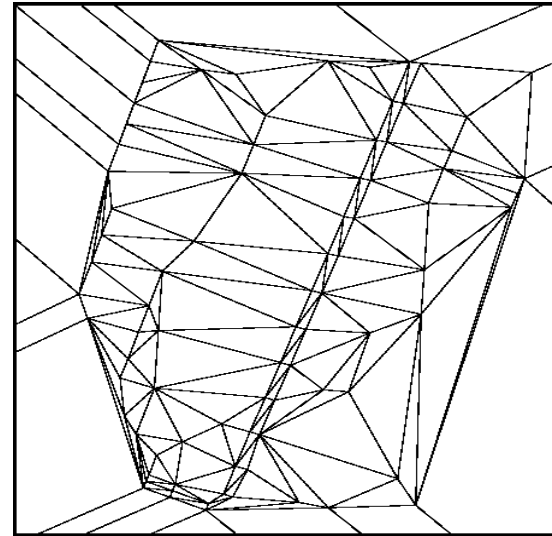
Local mapping functions

- Piecewise affine or cubic



Local mapping functions

- Piecewise affine or cubic



- Thin-Plate Splines (TPS)

$$\alpha_1 + \alpha_2 x + \alpha_3 y + \sum_{i=1}^N a_i g_i(\|x - x_i, y - y_i\|),$$

$$g_i(t) = t^2 \log t.$$

Airport images

Reference



Sensed (simulation)



From D. N. Fogel et al., UCSB

Airport images

Reference



Affine mapping



Airport images

Reference



Cubic mapping

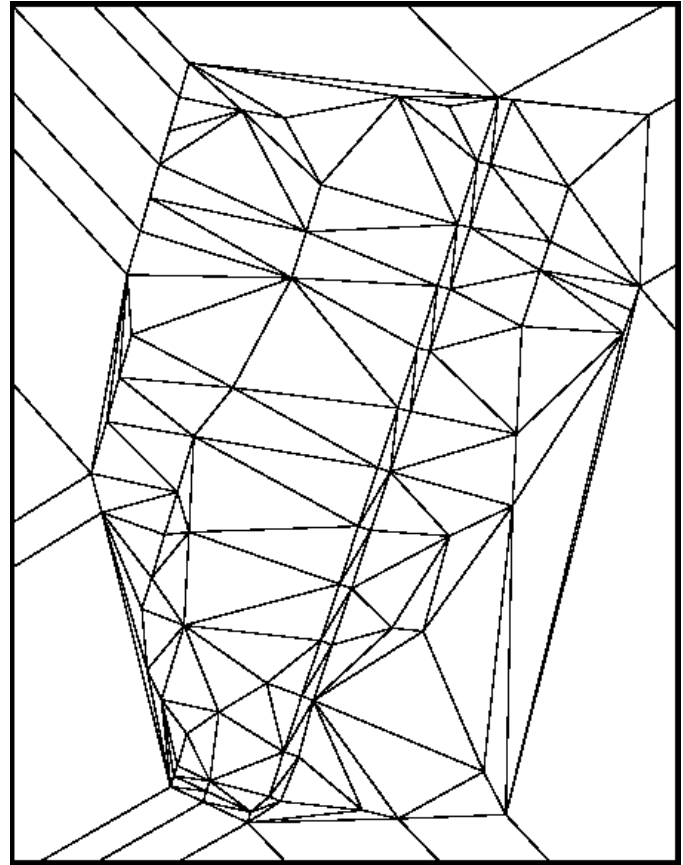


Airport images

Reference



Piecewise affine

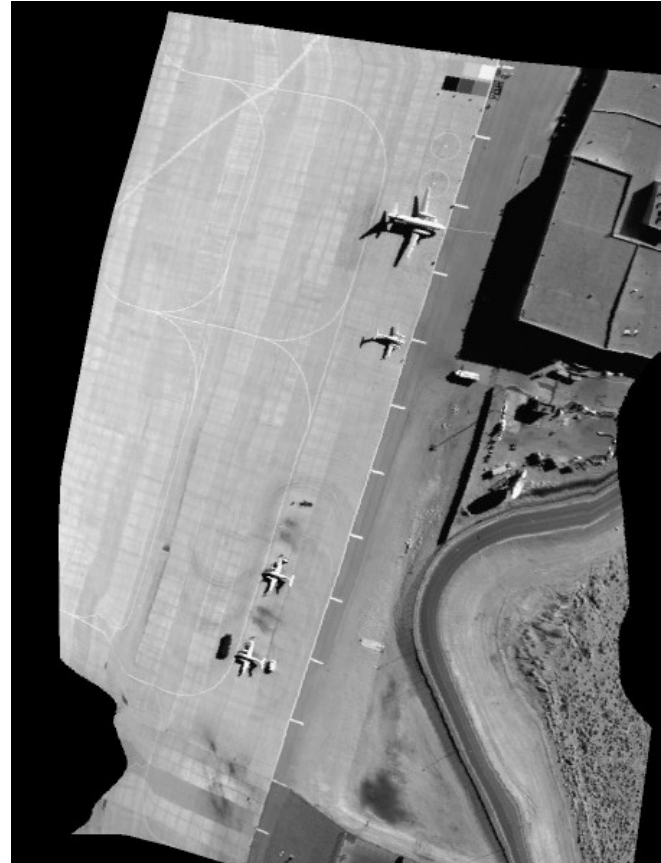


Airport images

Reference



Piecewise affine



Airport images

Reference



Thin-plate splines

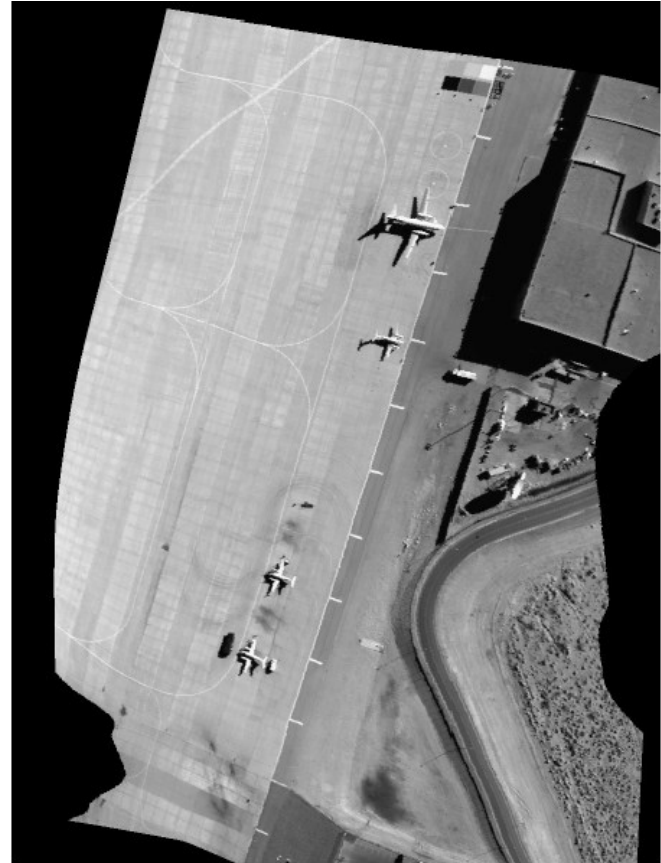


Image resampling

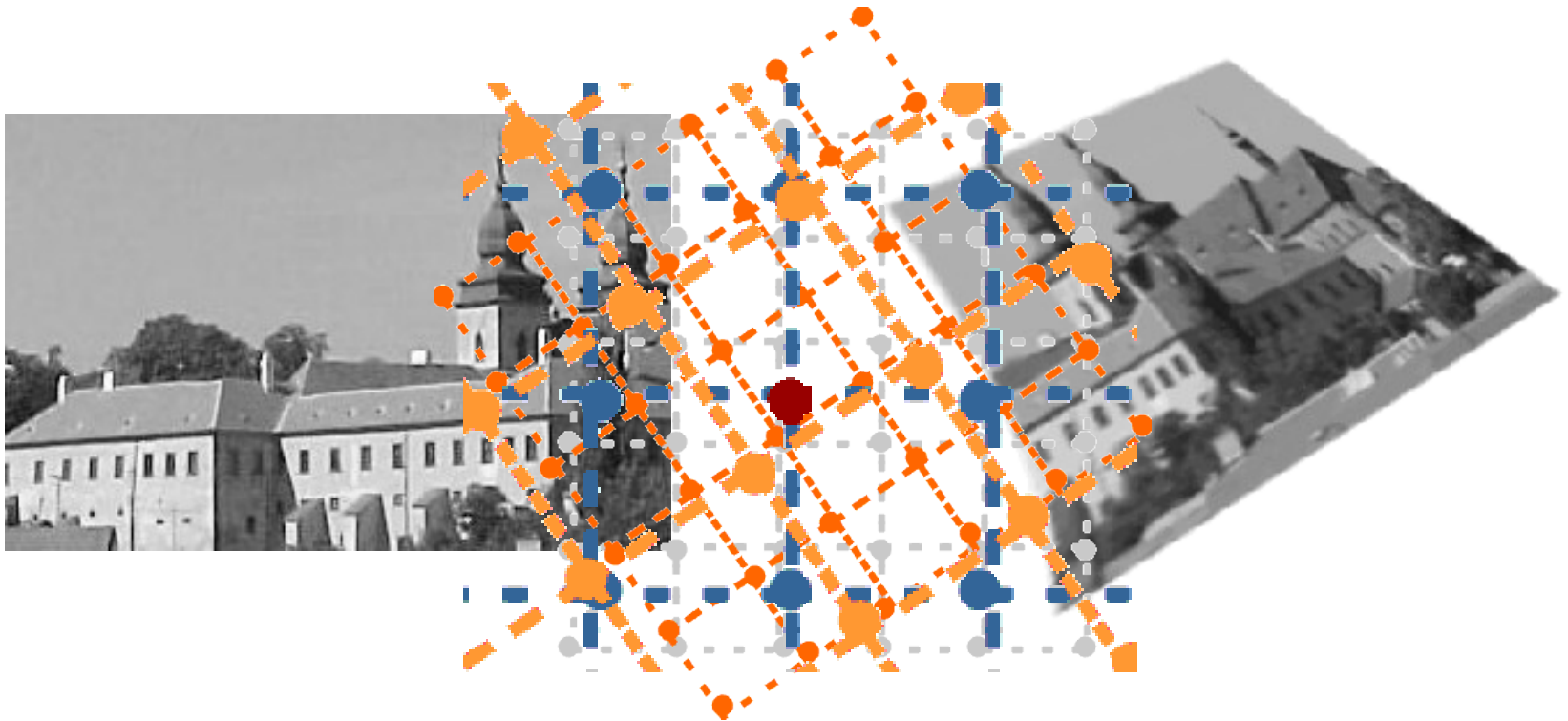


Image resampling

Forward x backward methods

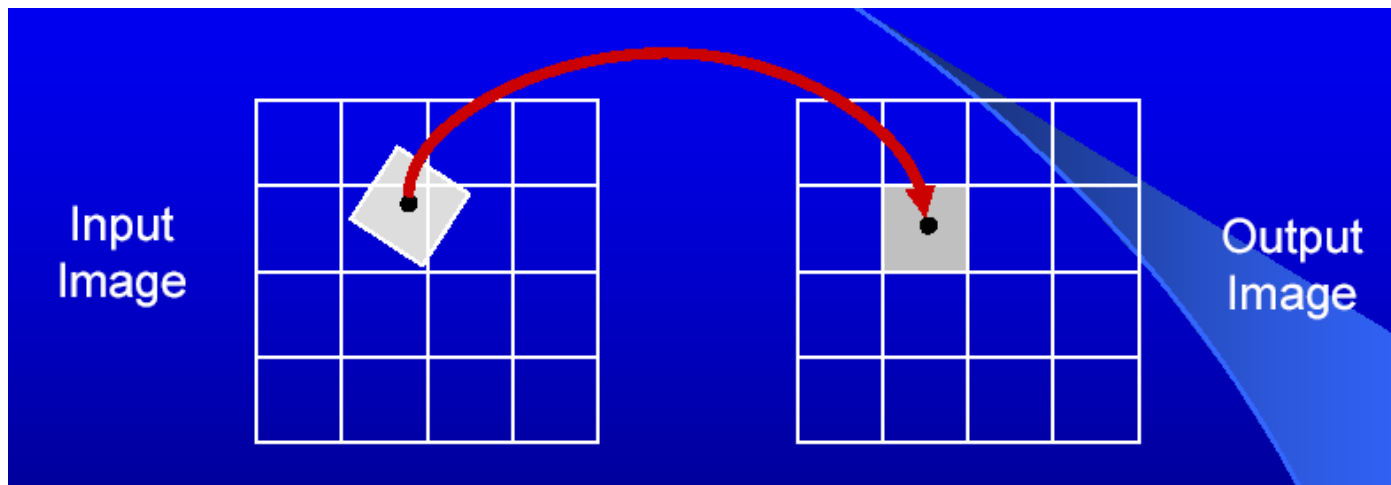
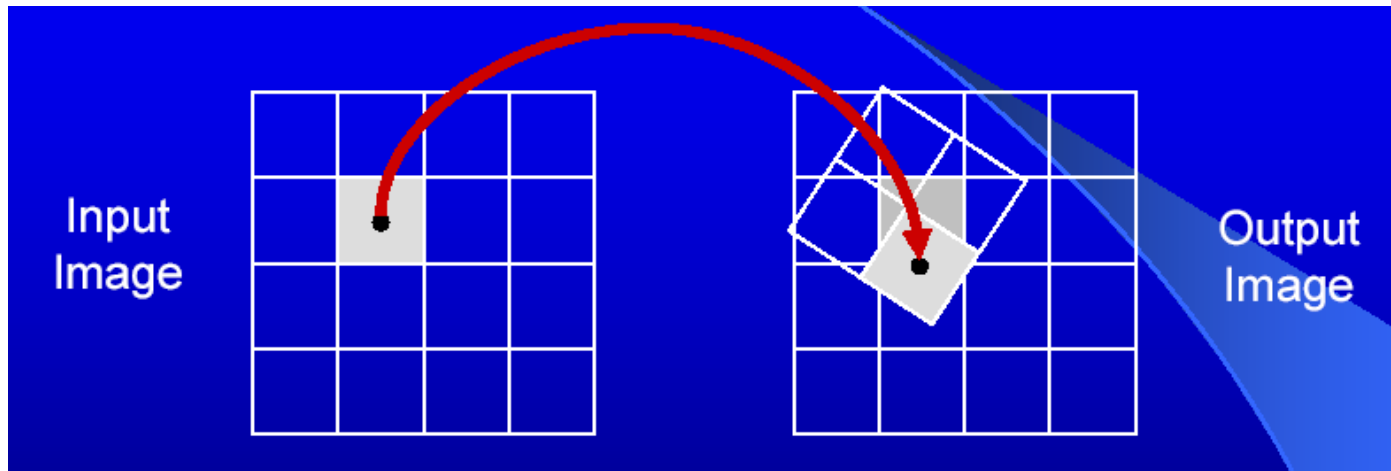


IMAGE RESAMPLING AND TRANSFORMATION

Interpolation nearest neighbor
 bilinear
 bicubic

Implementation 1-D convolution

$$f(x_o, k) = \sum d(I, k).c(i-x_o)$$

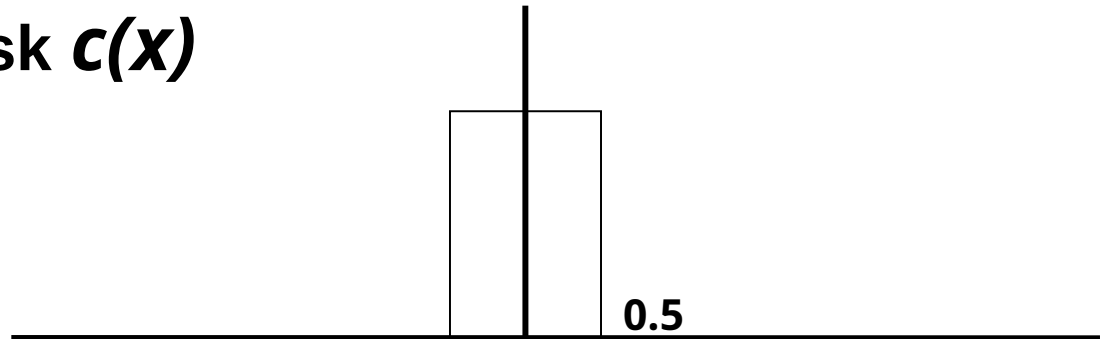
$$f(x_o, y_o) = \sum f(x_o, j).c(j-y_o)$$

ideal $c(x) = k.sinc(kx)$

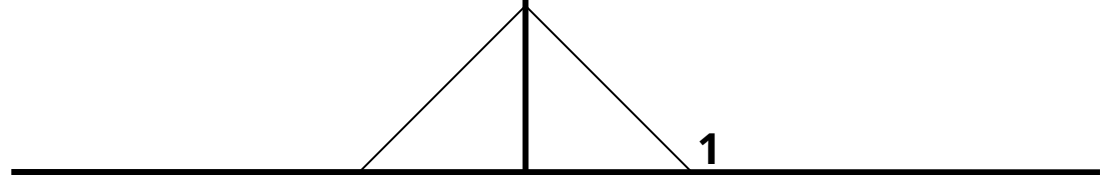
IMAGE RESAMPLING AND TRANSFORMATION

Interpolation mask $c(x)$

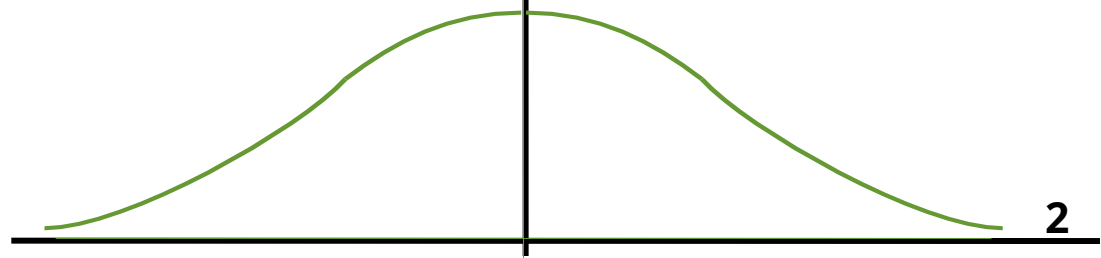
nearest neighbor



linear



smooth cubic



Interpolation methods

Original



Nearest
neighbor



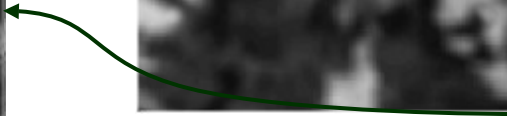
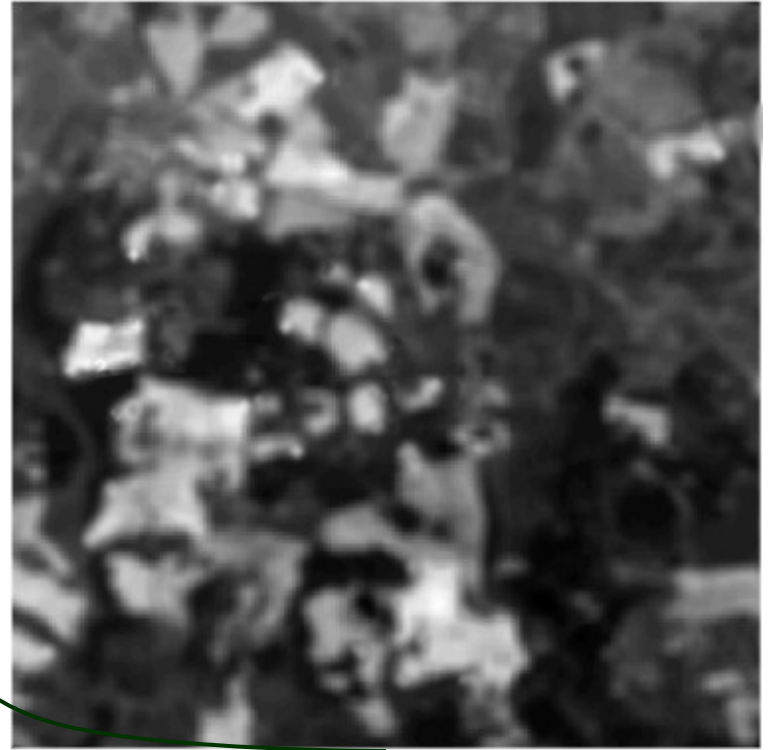
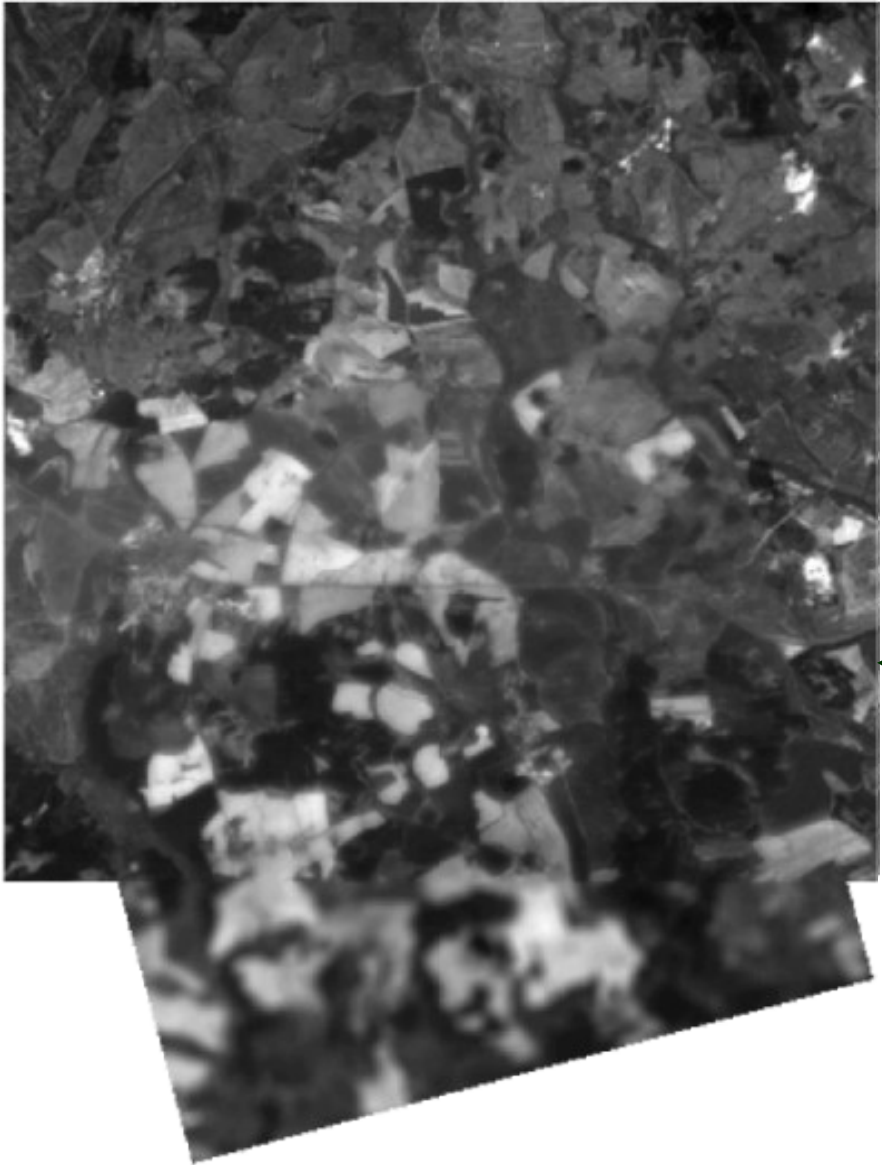
Bilinear



Bicubic



Registered images

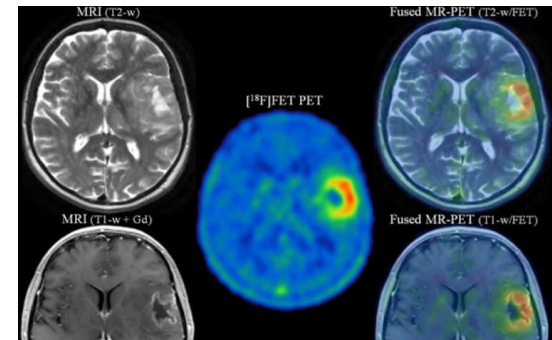
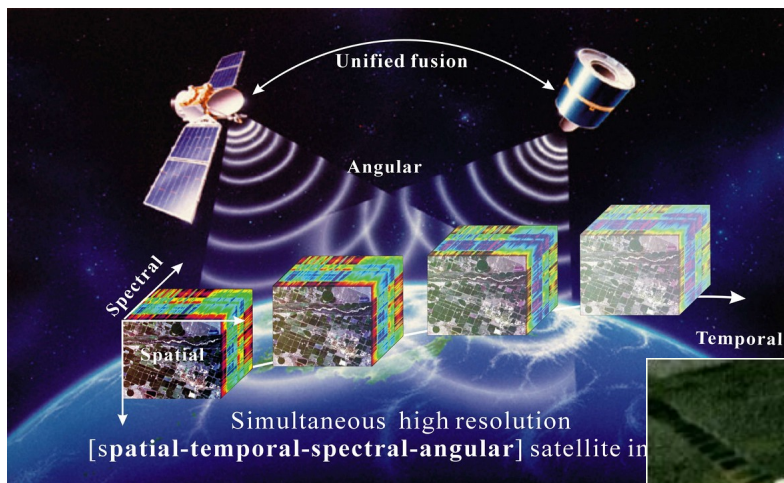


Applications

- Remote sensing
- Medicine
- Art history
- Robot vision
- Image warping
- Image fusion

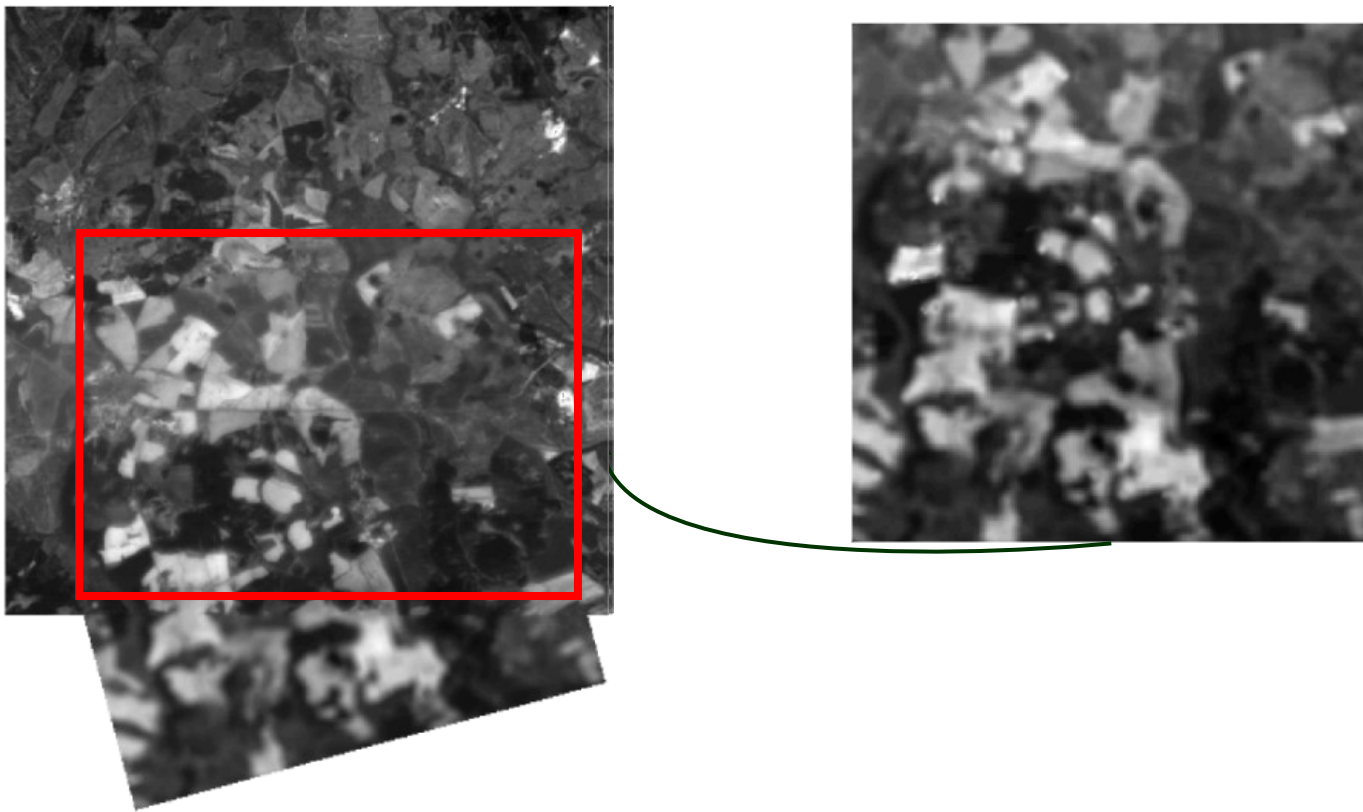
Image Fusion

Combining several images of the same scene into a single image of a higher quality



Two stages of image fusion

1. Image-to-image registration



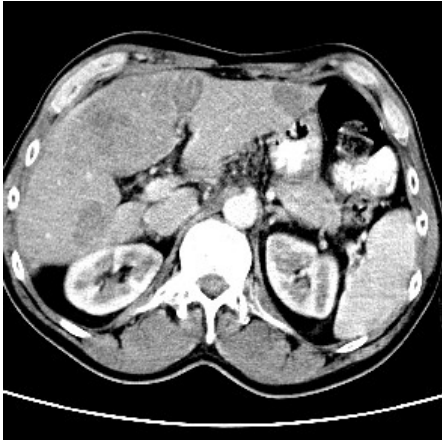
2. The fusion itself

Fusion levels

- **Pixel level**
- **Feature level**
- **Decision level**

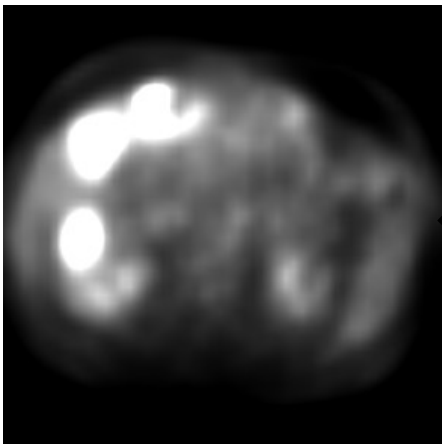
Pixel-level fusion

A



CT

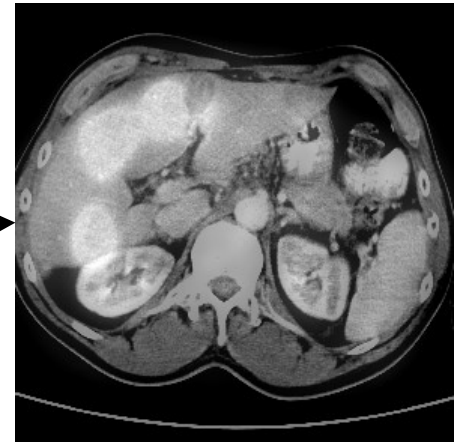
B



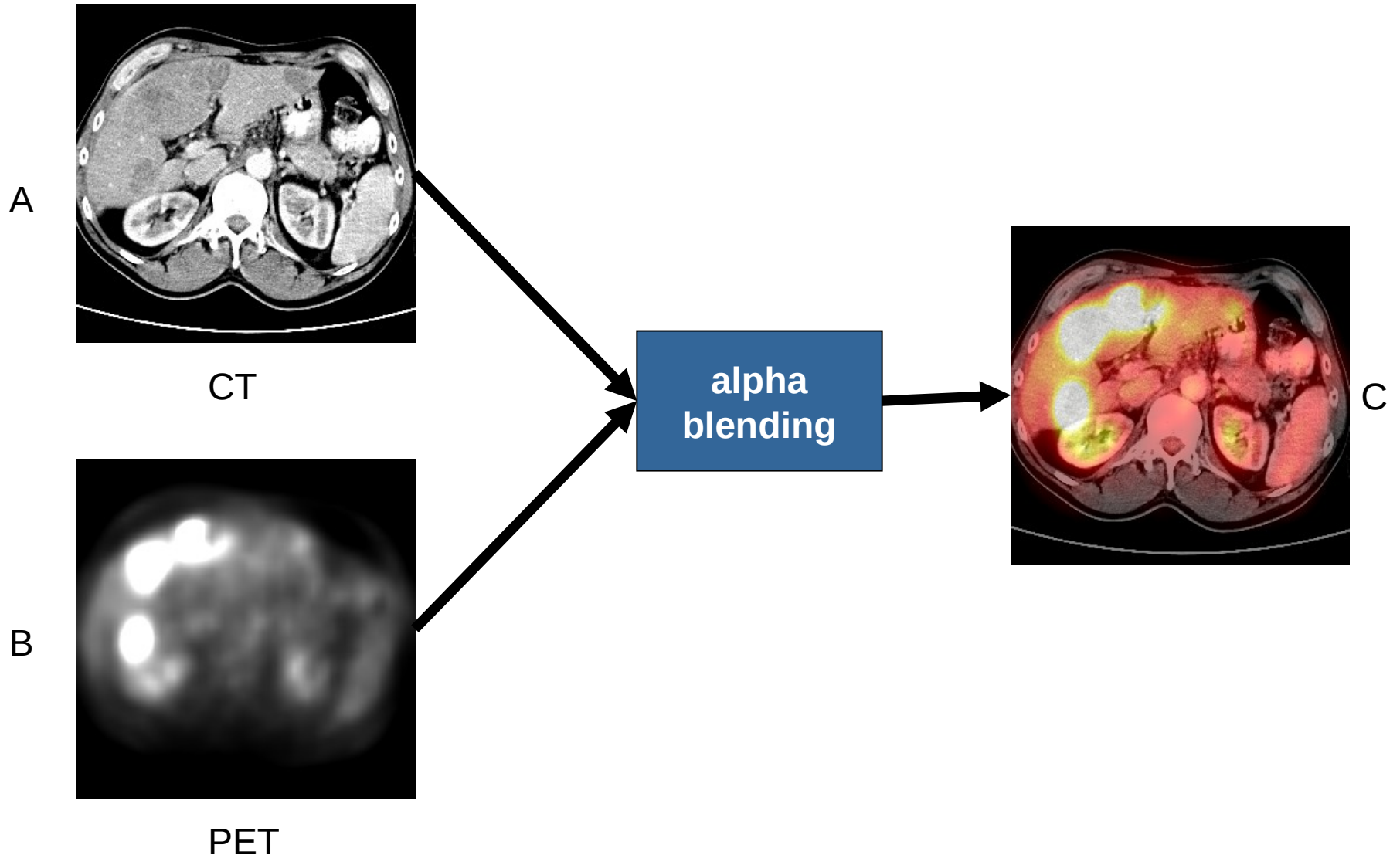
PET

weighted
averaging

C



Pixel-level fusion



Digital subtraction angiography

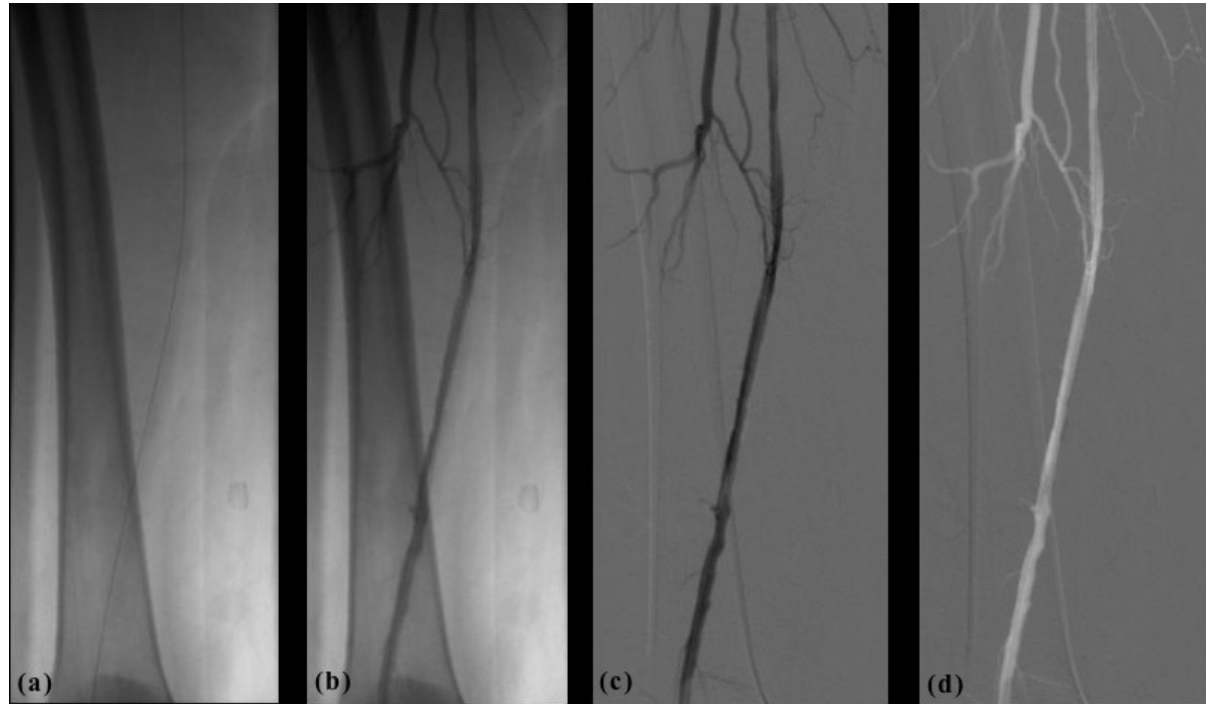


IMAGE REGISTRATION: ACCURACY EVALUATION

Localization error - displacement of features
- due to detection method

Matching error - false matches
- ensured by robust matching (hybrid)
- consistency check, cross-validation

Alignment error - difference between model and reality
- test point error



Thank you !

Any questions ?